

Matrix Decomposition Theorem

Matrix **decomposition** is a factorization of a matrix into some canonical form (Standard form/Normal form). There are many different matrix decompositions; each finds use among a particular class of problems.

Decompositions related to solving systems of linear equations

- LU decomposition
- Cholesky decomposition
- QR decomposition

Decompositions based on eigenvalues and related concepts

- **Eigen decomposition (also called spectral/diagonal decomposition)**
- Singular value decomposition

A. Eigen/diagonal Decomposition or spectral decomposition Theorem:

In linear algebra, eigendecomposition or sometimes spectral decomposition is the factorization of a matrix into a canonical form, whereby the matrix is represented in terms of its eigenvalues and eigenvectors. Only diagonalizable matrices can be factorized in this way.

Let A be the given Matrix, D is a diagonal matrix with the corresponding eigen values of Matrix A and **P be a matrix of eigenvectors** of a given Matrix A, Then A can be written as an **eigen decomposition** $A = PDP^{-1}$

✓ **Example 67:** Examine the eigen decomposition for the matrix $A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$

$$\text{Let } A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$$

$$\therefore A - \lambda I = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{pmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} = (4-\lambda)(4-\lambda) - 1$$

$$\Rightarrow |A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} = (4-\lambda)(4-\lambda) - 1 = 0$$

$$\Rightarrow (16 - 4\lambda - 4\lambda + \lambda^2) - 1 = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 16 - 1 = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 15 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda - 3\lambda + 15 = 0$$

$$\Rightarrow \lambda(\lambda - 5) - 3(\lambda - 5) = 0$$

$$\Rightarrow (\lambda - 5)(\lambda - 3) = 0$$

$$\Rightarrow \lambda = 5, \lambda = 3$$

$$\therefore D = \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}$$

Now to find out **the eigenvector** of the matrix A,

Let a non-zero **vector** $\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$

We have, $A\mathbf{v} = \lambda\mathbf{v}$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4x + y \\ x + 4y \end{pmatrix} = 3 \begin{pmatrix} x \\ y \end{pmatrix} \quad [\text{For } \lambda = 3]$$

$$\Rightarrow \begin{pmatrix} 4x + y \\ x + 4y \end{pmatrix} = \begin{pmatrix} 3x \\ 3y \end{pmatrix}$$

$$\Rightarrow 4x + y = 3x$$

$$x + 4y = 3y$$

$$\Rightarrow x + y = 0$$

$$x + y = 0$$

$$\Rightarrow x + y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in **particular** $(2-1) = 1$ **free variable**, which is y.

Let $y = 1$ then $x = -1$

$$\therefore \text{Eigenvector } \mathbf{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \text{ for } \lambda = 3$$

Again for $\lambda = 5$

Let a non-zero vector $\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$

We have, $A\mathbf{v} = \lambda\mathbf{v}$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 5 \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4x + y \\ x + 4y \end{pmatrix} = 5 \begin{pmatrix} x \\ y \end{pmatrix} \quad [\text{For } \lambda = 5]$$

$$\Rightarrow \begin{pmatrix} 4x + y \\ x + 4y \end{pmatrix} = \begin{pmatrix} 5x \\ 5y \end{pmatrix}$$

$$\Rightarrow 4x + y = 5x$$

$$x + 4y = 5y$$

$$\Rightarrow -x + y = 0$$

$$x - y = 0$$

$$\Rightarrow x - y = 0$$

$$x - y = 0$$

$$\Rightarrow \cancel{x} - y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in **particular (2-1) = 1 free variable**, which is y.

Let $y=1$, then $x=1$

$$\therefore \text{Eigenvector } \mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ for } \lambda = 5$$

P be a matrix of eigenvectors of a given Matrix A.

$$\therefore \mathbf{P} = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$$

Now to find out $\mathbf{P}^{-1}$

$$c_{11} = 1, c_{12} = -1, c_{21} = -1, c_{22} = -1$$

$$\mathbf{P}^c = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$\text{Adj } \mathbf{P} = (\mathbf{P}^c)^T = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$|\mathbf{P}| = \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} = -1 - 1 = -2$$

$$\therefore \mathbf{P}^{-1} = \frac{\text{Adj } \mathbf{P}}{|\mathbf{P}|} = -\frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

Then A can be written as an eigen decomposition $\mathbf{A} = \mathbf{PDP}^{-1}$

Justification:

$$\begin{aligned} \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} &= -\frac{1}{2} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix} \\ &= -\frac{1}{2} \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 & -3 \\ -5 & -5 \end{pmatrix} \\ &= -\frac{1}{2} \begin{pmatrix} -3-5 & 3-5 \\ 3-5 & -3-5 \end{pmatrix} \end{aligned}$$

$$= -\frac{1}{2} \begin{pmatrix} -8 & -2 \\ -2 & -8 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} = A \quad (\text{Proved})$$

$$A = PDP^{-1} \quad (\text{Proved})$$

Home task: Find eigen decomposition of the matrix $= \begin{pmatrix} 7 & 3 \\ 3 & -1 \end{pmatrix}$ and $= \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$

Q-01: What is the difference between Eigen decomposition and SVD?

The SVD always exists for any sort of rectangular or square matrix, whereas the eigen decomposition can only exist for square matrices, and even among square matrices sometimes it doesn't exist.

What is singular value of a Matrix?

The singular values of a $m \times n$ matrix A are the square roots of the eigenvalues of the symmetric $n \times n$ matrix $A^T A$. The Eigen values of $A^T A$ are

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > \lambda_{r+1} = \dots = \lambda_n = 0$$

Since large eigenvalues carry significant representation of the matrix, we arrange the values of the Eigen Values from the largest to the smallest and drop/ignore the values towards the last since small eigenvalues are less indicator.

$\sigma_i = \sqrt{\lambda_i}$ are called the singular value of A where $i = 1, 2, \dots, n$.

It is customary to denote the singular values by $\sigma_1, \sigma_2, \sigma_3, \dots, \sigma_n$ and to list them in decreasing order:

$$\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \dots \geq \sigma_n$$

Singular Value Decomposition:

In linear algebra, the **singular value decomposition (SVD)** is an important factorization of a rectangular real or complex matrix, with several applications **in signal processing, image processing and compression, gene expression analysis and statistics**. Applications which employ the SVD include computing the pseudo-inverse, matrix approximation, and determining the rank, range and null space of a matrix.

The spectral theorem says that normal matrices, which by definition must be square, can be unitarily diagonalized using a basis of eigenvectors. The SVD can be seen as a generalization of the spectral theorem to arbitrary, not necessarily square, matrices.

The singular value decomposition (SVD) is a generalization of the eigen-decomposition, which can be used to analyze rectangular matrices (the eigen-decomposition is defined only for squared matrices). By analogy with the eigen-decomposition, which decomposes

a matrix into two simple matrices, the main idea of the SVD is to decompose a rectangular matrix into three simple matrices: Two orthogonal matrices and one diagonal matrix.

Matrix A is decomposed using SVD to acquire an orthogonal base in the multidimensional feature space:

$$\underbrace{\begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix}}_A = \underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_U \underbrace{\begin{bmatrix} * \\ * \end{bmatrix}}_S \underbrace{\begin{bmatrix} * & * \\ * & * \end{bmatrix}}_{V^T}$$

$$A = USV^T$$

Here,

A → Given Matrix

U → Matrix of Eigen Vectors of the Matrix AA^T

V → Matrix of Eigen Vectors of the Matrix $A^T A$

S → Roots of Eigen Values of the Matrix AA^T or $A^T A$

V^T → Transpose of the Matrix **V**

Example 68: Compute Singular Value Decomposition (SVD) for the given matrix

$$\begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$$

Answer: Let, $A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

In order to find U, Now we need to find the eigenvectors for AA^T

Given, $A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

$$\therefore A^T = \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$$

So, $AA^T = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$

$$= \begin{pmatrix} 4+4 & 2-2 \\ 2-2 & 1+1 \end{pmatrix}$$

$$= \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}$$

------(i)

Now we need to find the eigenvalues of AA^T

$$\begin{aligned}
\mathbf{A}\mathbf{A}^T - \lambda\mathbf{I} &= \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \\
&= \begin{pmatrix} 8-\lambda & 0 \\ 0 & 2-\lambda \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
|\mathbf{A}\mathbf{A}^T - \lambda\mathbf{I}| &= 0 \\
\Rightarrow \begin{vmatrix} 8-\lambda & 0 \\ 0 & 2-\lambda \end{vmatrix} &= 0 \\
\Rightarrow (8-\lambda)(2-\lambda) &= 0 \\
\Rightarrow (\lambda-8)(\lambda-2) &= 0 \\
\Rightarrow \lambda &= 8, 2
\end{aligned}$$

Therefore eigenvalues are 8 and 2. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$\mathbf{D} = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

$$\text{The matrix of singular values } \mathbf{S} = \sqrt{\mathbf{D}} = \begin{pmatrix} \sqrt{8} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$$

$$\mathbf{S} = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \text{-----(ii)}$$

Now we need to find the eigenvector of $\mathbf{A}\mathbf{A}^T$

We have, $\mathbf{A}\mathbf{u}_1 = \lambda\mathbf{u}_1$

$$[\mathbf{A}\mathbf{V} = \lambda\mathbf{V}]$$

$$\text{Let } \mathbf{u}_1 = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\therefore \mathbf{A}\mathbf{A}^T\mathbf{u}_1 = \lambda\mathbf{u}_1$$

$$\begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 8 \begin{pmatrix} x \\ y \end{pmatrix}$$

[For $\lambda = 8$ & from (i) putting the value of $\mathbf{A}\mathbf{A}^T$]

$$\Rightarrow \begin{pmatrix} 8x \\ 2y \end{pmatrix} = \begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$$\Rightarrow 8x = 8x$$

$$2y = 8y$$

$$\Rightarrow 8x - 8x = 0$$

$$2y - 8y = 0$$

$$\Rightarrow 8x = 8x$$

$$-6y = 0$$

$$\Rightarrow 8x = 8x$$

$$y = 0$$

$$\therefore y = 0$$

$$\text{Let } x = -1$$

$$\therefore \mathbf{u}_1 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \text{ It is a unit vector or normal vector. Since the length of } \mathbf{u}_1$$

$$\text{is } |\mathbf{u}_1| = \sqrt{(-1)^2 + (0)^2} = \sqrt{1} = 1$$

$$\text{Again, Let } \mathbf{u}_2 = \begin{pmatrix} x \\ y \end{pmatrix}$$

Here,

$$\therefore \mathbf{A}\mathbf{A}^T \mathbf{u}_2 = \lambda \mathbf{u}_2$$

$$\begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2 \begin{pmatrix} x \\ y \end{pmatrix} \quad [\text{For } \lambda = 2 \text{ \& from (i) putting the value of } \mathbf{A}\mathbf{A}^T]$$

$$\Rightarrow \begin{pmatrix} 8x \\ 2y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\Rightarrow 8x = 2x$$

$$2y = 2y$$

$$\Rightarrow 8x - 2x = 0$$

$$2y = 2y$$

$$\Rightarrow 6x = 0$$

$$2y = 2y$$

$$\therefore x = 0$$

$$2y = 2y$$

$$\text{Let } y = 1$$

$$\therefore \mathbf{u}_2 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ It is a unit vector or normal vector. Since the length of } \mathbf{u}_1$$

$$\text{is } |\mathbf{u}_2| = \sqrt{(0)^2 + (1)^2} = \sqrt{1} = 1$$

$$\therefore U = (u_1 \quad u_2) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \text{-----(iii)}$$

Now we need to find the eigenvectors for $A^T A$

$$\text{Given, } A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$$

$$\therefore A^T = \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$$

$$\begin{aligned} \therefore A^T A &= \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \text{-----(iv)} \end{aligned}$$

Now we need to find the eigenvalues of $A^T A$

$$\begin{aligned} \therefore A^T A - \lambda I &= \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} 5-\lambda & -3 \\ -3 & 5-\lambda \end{pmatrix} \end{aligned}$$

$$\therefore |A^T A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 5-\lambda & -3 \\ -3 & 5-\lambda \end{vmatrix} = 0$$

$$(5-\lambda)(5-\lambda) - 9 = 0$$

$$\Rightarrow 25 - 10\lambda + \lambda^2 - 9 = 0$$

$$\Rightarrow \lambda^2 - 10\lambda + 16 = 0$$

$$\Rightarrow (\lambda - 8)(\lambda - 2) = 0$$

$$\Rightarrow \lambda = 8, 2$$

Therefore eigenvalues are 8 and 2. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$D = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

$$\text{The matrix of singular values } S = \begin{pmatrix} \sqrt{8} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$$

$$S = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \text{-----(v)}$$

Now we need find the eigenvectors of $A^T A$ which are the columns of V .

We have, $Av_1 = \lambda v_1$

$$[AV = \lambda V]$$

$$\text{Let } v_1 = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\therefore A^T A v_1 = \lambda v_1$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 8 \begin{pmatrix} x \\ y \end{pmatrix}$$

[For $\lambda = 8$ & from (iv) putting the value of $A^T A$]

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$$\begin{pmatrix} 5x - 3y \\ -3x + 5y \end{pmatrix} = \begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$$\Rightarrow 5x - 3y = 8x$$

$$-3x + 5y = 8y$$

$$\Rightarrow -3x - 3y = 0$$

$$-3x - 3y = 0$$

$$\Rightarrow x + y = 0$$

$$x + y = 0$$

$$\Rightarrow \cancel{x} + y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is y .

Let $y = 1 \therefore x = -1$

$$\therefore v_1 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}; \text{ It is not unit vector or normal vector. Since the length of } v_1$$

is $|v_1| = \sqrt{(-1)^2 + (1)^2} = \sqrt{2}$; this vector v_1 has to be converted into unit vector by dividing its magnitude or its length.

$$\text{We have, } v_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$\therefore \mathbf{v}_1 = \begin{pmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

[Dividing by its magnitude or its length $\sqrt{2}$]

$$\mathbf{v}_1 = \begin{pmatrix} \frac{-\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$$

[Multiplying by $\sqrt{2}$ on numerator and denominator]

Similarly,

We have, $\mathbf{A}\mathbf{v}_2 = \lambda\mathbf{v}_2$

$$[\mathbf{A}\mathbf{V} = \lambda\mathbf{V}]$$

$$\text{Let } \mathbf{v}_2 = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\therefore \mathbf{A}^T \mathbf{A} \mathbf{v}_2 = \lambda \mathbf{v}_2$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2 \begin{pmatrix} x \\ y \end{pmatrix}$$

[For $\lambda = 2$ & from (iv) putting the value of $\mathbf{A}^T \mathbf{A}$]

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\begin{pmatrix} 5x - 3y \\ -3x + 5y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\Rightarrow 3x - 3y = 0$$

$$-3x + 3y = 0$$

$$\Rightarrow x - y = 0$$

$$x - y = 0$$

$$\Rightarrow x - y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular $(2-1)=1$ free variable, which is y .

Let $y = 1 \therefore x = 1$

$$\therefore \mathbf{v}_2 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ It is not unit vector or normal vector. Since the length of } \mathbf{v}_2$$

is $|\mathbf{v}_2| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$; this vector $\mathbf{v}_2$ has to be converted into unit vector by dividing its magnitude or its length.

We have,

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\text{Thus, } \mathbf{v}_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} \quad [\text{Dividing by its magnitude or its length } \sqrt{2}]$$

$$\Rightarrow \mathbf{v}_2 = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix} \quad [\text{Multiplying by } \sqrt{2} \text{ on numerator and denominator}]$$

Now we need to construct the augmented orthogonal matrix V,

$$\mathbf{V} = (\mathbf{v}_1 \quad \mathbf{v}_2) = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\therefore \mathbf{V}^T = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \quad \text{-----(vi)}$$

From (ii), (iii) & (vi), Putting the values of $\mathbf{U}, \mathbf{S}, \mathbf{V}^T$ in the relation $\mathbf{A} = \mathbf{USV}^T$

Therefore $\mathbf{A} = \mathbf{USV}^T$

$$\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \text{Answer}$$

Justification Test:

$$\mathbf{A} = \mathbf{USV}^T$$

$$\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \quad [\text{Putting the value of } \mathbf{U}, \mathbf{V}^T, \mathbf{S}]$$

$$\mathbf{A} = \mathbf{USV}^T$$

$$\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -2\sqrt{2} \times \frac{\sqrt{2}}{2} + 0 & 2\sqrt{2} \times \frac{\sqrt{2}}{2} + 0 \\ 0 + \sqrt{2} \times \frac{\sqrt{2}}{2} & 0 + \sqrt{2} \times \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -2 & 2 \\ 1 & 1 \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} -1 \times (-2) + 0 & (-1) \times 2 + 0 \\ 0 + 1 & 0 + 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix} \text{ [Given Matrix] Proved}$$

Example 69: Compute Singular Value Decomposition (SVD) for the given matrix

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

In order to find U, In order to find U, Now we need to find the eigenvectors for AA^T
Given,

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

$$\therefore A^T = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$$

$$\text{So, } AA^T = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix} \quad \text{-----(i)}$$

Now we need find the eigenvalues of AA^T

$$|AA^T - \lambda I| = 0$$

$$\left| \begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right| = 0$$

$$\left| \begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right| = 0$$

Therefore,

$$\begin{vmatrix} 11-\lambda & 1-0 \\ 1-0 & 11-\lambda \end{vmatrix} = 0$$

$$(11-\lambda)(11-\lambda) - 1 = 0$$

$$\Rightarrow 121 - 11\lambda - 11\lambda + \lambda^2 - 1 = 0$$

$$\Rightarrow \lambda^2 - 22\lambda + 121 - 1 = 0$$

$$\Rightarrow \lambda^2 - 22\lambda + 120 = 0$$

$$\Rightarrow \lambda^2 - 10\lambda - 12\lambda + 120 = 0$$

$$\Rightarrow (\lambda - 10)(\lambda - 12) = 0$$

$$\Rightarrow \lambda = 10, 12$$

Therefore eigenvalues are 10 and 12. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$D = \begin{pmatrix} 12 & 0 \\ 0 & 10 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

Since large eigenvalues carry significant representation of the matrix, we arrange the values of the Eigen Values from the largest to the smallest and drop/ignore the values towards the last since small eigenvalues are less indicator.

What is the use of SVD?

01. SVD can be used to compute optimal low-rank approximations of arbitrary matrices.
02. Face recognition: represent the face images as eigenfaces and compute distance between the query face image in the principal component space.

B. LU Decomposition

A procedure for decomposing a $N \times N$ matrix A into a product of a lower triangular matrix L and an upper triangular matrix U,

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{pmatrix}$$

LU decomposition is implemented in *Mathematica* as LU Decomposition[m].

Example 70

Solve the linear system of equations using LU decomposition

$$u_1 + 2u_2 + 3u_3 = 5$$

$$2u_1 - 4u_2 + 6u_3 = 18$$

$$3u_1 - 9u_2 - 3u_3 = 6$$

Solution:

Given the linear system of equations

$$u_1 + 2u_2 + 3u_3 = 5$$

$$2u_1 - 4u_2 + 6u_3 = 18$$

$$3u_1 - 9u_2 - 3u_3 = 6$$

This can be written as

$$\mathbf{Au} = \mathbf{b} \quad \text{-----(i)}$$

Where,

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix} \quad \mathbf{u} = \begin{pmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \\ \mathbf{u}_3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

Again, According to LU decomposition,

$$\mathbf{A} = \mathbf{LU} \quad \text{-----(ii)}$$

Where,

$$\mathbf{L} = \begin{pmatrix} \mathbf{l}_{11} & 0 & 0 \\ \mathbf{l}_{21} & \mathbf{l}_{22} & 0 \\ \mathbf{l}_{31} & \mathbf{l}_{32} & \mathbf{l}_{33} \end{pmatrix} \quad \text{and} \quad \mathbf{U} = \begin{pmatrix} 1 & \mathbf{u}_{12} & \mathbf{u}_{13} \\ 0 & 1 & \mathbf{u}_{23} \\ 0 & 0 & 1 \end{pmatrix}$$

From (ii), $\mathbf{A} = \mathbf{LU}$

$$\Rightarrow \mathbf{LU} = \mathbf{A}$$

$$\begin{pmatrix} \mathbf{l}_{11} & 0 & 0 \\ \mathbf{l}_{21} & \mathbf{l}_{22} & 0 \\ \mathbf{l}_{31} & \mathbf{l}_{32} & \mathbf{l}_{33} \end{pmatrix} \begin{pmatrix} 1 & \mathbf{u}_{12} & \mathbf{u}_{13} \\ 0 & 1 & \mathbf{u}_{23} \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix}$$

$$\begin{pmatrix} \mathbf{l}_{11} + 0 + 0 & \mathbf{l}_{11}\mathbf{u}_{12} + 0 + 0 & \mathbf{l}_{11}\mathbf{u}_{13} + 0 + 0 \\ \mathbf{l}_{21} + 0 + 0 & \mathbf{l}_{21}\mathbf{u}_{12} + \mathbf{l}_{22} + 0 & \mathbf{l}_{21}\mathbf{u}_{13} + \mathbf{l}_{22}\mathbf{u}_{23} + 0 \\ \mathbf{l}_{31} + 0 + 0 & \mathbf{l}_{31}\mathbf{u}_{12} + \mathbf{l}_{32} + 0 & \mathbf{l}_{31}\mathbf{u}_{13} + \mathbf{l}_{32}\mathbf{u}_{23} + \mathbf{l}_{33} \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -4 & 6 \\ 3 & -9 & -3 \end{pmatrix}$$

Now,

We can write

$$\mathbf{l}_{11} = 1 \quad \text{-----(iii)}$$

$$\mathbf{l}_{11}\mathbf{u}_{12} = 2 \quad \text{-----(iv)}$$

$$\mathbf{l}_{11}\mathbf{u}_{13} = 3 \quad \text{-----(v)}$$

$$\mathbf{l}_{21} = 2 \quad \text{-----(vi)}$$

$$\mathbf{l}_{21}\mathbf{u}_{12} + \mathbf{l}_{22} = -4 \quad \text{-----(vii)}$$

$$\mathbf{l}_{21}\mathbf{u}_{13} + \mathbf{l}_{22}\mathbf{u}_{23} = 6 \quad \text{-----(viii)}$$

$$\mathbf{l}_{31} = 3 \quad \text{-----(ix)}$$

$$\mathbf{l}_{31}\mathbf{u}_{12} + \mathbf{l}_{32} = -9 \quad \text{-----(x)}$$

$$\mathbf{l}_{31}\mathbf{u}_{13} + \mathbf{l}_{32}\mathbf{u}_{23} + \mathbf{l}_{33} = -3 \quad \text{-----(xi)}$$

From (iii) & (iv)

$$l_{11} = 1$$

$$l_{11}u_{12} = 2$$

$$1.u_{12} = 2$$

$$u_{12} = 2 \quad \text{-----(xii)}$$

From (iii) & (v)

$$l_{11}u_{13} = 3$$

$$1.u_{13} = 3$$

$$u_{13} = 3 \quad \text{-----(xiii)}$$

From (vii),

$$l_{21}u_{12} + l_{22} = -4$$

$$2.2 + l_{22} = -4 \quad [u_{12} = 2 ; l_{21} = 2]$$

$$4 + l_{22} = -4$$

$$\therefore l_{22} = -8 \quad \text{-----(xiv)}$$

From (viii),

$$l_{21}u_{13} + l_{22}u_{23} = 6$$

$$2.3 - 8u_{23} = 6 \quad [l_{21} = 2 ; u_{13} = 3 ; \therefore l_{22} = -8]$$

$$-8u_{23} = 6 - 6$$

$$u_{23} = 0 \quad \text{-----(xv)}$$

From (x),

$$l_{31}u_{12} + l_{32} = -9$$

$$3.2 + l_{32} = -9 \quad [l_{31} = 3 ; u_{12} = 2]$$

$$l_{32} = -9 - 6$$

$$l_{32} = -15 \quad \text{-----(xvi)}$$

From (xi)

$$l_{31}u_{13} + l_{32}u_{23} + l_{33} = -3$$

$$3.3 - 15.0 + l_{33} = -3 \quad [l_{31} = 3 ; u_{13} = 3 ; l_{32} = -15 ; u_{23} = 0]$$

$$l_{33} = -3 - 9$$

$$l_{33} = -12 \quad \text{-----(xvii)}$$

$$\therefore \mathbf{L} = \begin{pmatrix} l_{11} & \mathbf{0} & \mathbf{0} \\ l_{21} & l_{22} & \mathbf{0} \\ l_{31} & l_{32} & l_{33} \end{pmatrix}$$

$$\Rightarrow \mathbf{L} = \begin{pmatrix} 1 & \mathbf{0} & \mathbf{0} \\ 2 & -8 & \mathbf{0} \\ 3 & -15 & -12 \end{pmatrix}$$

And

$$\mathbf{U} = \begin{pmatrix} 1 & \mathbf{u}_{12} & \mathbf{u}_{13} \\ \mathbf{0} & 1 & \mathbf{u}_{23} \\ \mathbf{0} & \mathbf{0} & 1 \end{pmatrix}$$

$$\mathbf{U} = \begin{pmatrix} 1 & 2 & 3 \\ \mathbf{0} & 1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & 1 \end{pmatrix}$$

From (i),

$$\mathbf{A}\mathbf{u} = \mathbf{b}$$

$$\mathbf{LU}\mathbf{u} = \mathbf{b} \quad \text{[from (ii), } \mathbf{A} = \mathbf{LU}]$$

$$\mathbf{LU}\mathbf{u} = \mathbf{b} \quad \text{-----(xviii)}$$

Let,

$$\mathbf{U}\mathbf{u} = \mathbf{v} \quad \text{-----(xix)}$$

Where,

$$\text{Where, } \mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

From (xviii),

$$\mathbf{LU}\mathbf{u} = \mathbf{b}$$

$$\mathbf{L}\mathbf{v} = \mathbf{b}$$

$$\Rightarrow \begin{pmatrix} 1 & \mathbf{0} & \mathbf{0} \\ 2 & -8 & \mathbf{0} \\ 3 & -15 & -12 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 \times v_1 + 0 \times v_2 + 0 \times v_3 \\ 2 \times v_1 - 8 \times v_2 + 0 \times v_3 \\ 3 \times v_1 - 15 \times v_2 - 12 \times v_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} v_1 \\ 2v_1 - 8v_2 \\ 3v_1 - 15v_2 - 12v_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

$$v_1 = 5 \quad \text{-----(xx)}$$

$$2v_1 - 8v_2 = 18 \quad \text{-----(xxi)}$$

$$3v_1 - 15v_2 - 12v_3 = 6 \quad \text{-----(xxii)}$$

From (xxi),

$$2v_1 - 8v_2 = 18$$

$$2.5 - 8v_2 = 18 \quad [v_1 = 5]$$

$$10 - 8v_2 = 18$$

$$-8v_2 = 18 - 10$$

$$-8v_2 = 8$$

$$\therefore v_2 = -1 \quad \text{-----(xxiii)}$$

From (xxii),

$$3v_1 - 15v_2 - 12v_3 = 6$$

$$3.5 - 15(-1) - 12v_3 = 6 \quad [v_1 = 5; v_2 = -1]$$

$$15 + 15 - 12v_3 = 6$$

$$30 - 12v_3 = 6$$

$$-12v_3 = 6 - 30$$

$$-12v_3 = -24$$

$$\therefore v_3 = 2 \quad \text{-----(xxiv)}$$

From (xix),

$$Uu = v$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 \times u_1 + 2 \times u_2 + 3 \times u_3 \\ 0 \times u_1 + 1 \times u_2 + 0 \times u_3 \\ 0 \times u_1 + 0 \times u_2 + 1 \times u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} u_1 + 2u_2 + 3u_3 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$$u_1 + 2u_2 + 3u_3 = 5 \quad \text{-----(xxv)}$$

$$u_2 = -1 \quad \text{-----(xxvi)}$$

$$u_3 = 2 \quad \text{-----(xxvii)}$$

From (xxv),

$$u_1 + 2u_2 + 3u_3 = 5$$

$$u_1 + 2(-1) + 3 \cdot 2 = 5$$

$$u_1 - 2 + 6 = 5$$

$$u_1 = 5 - 4$$

$$u_1 = 1$$

$$u_1 = 1, u_2 = -1, u_3 = 2 \text{ Answer}$$

Example 71: Given the following system of linear equations, determine the value of each of the variables using the LU decomposition method.

$$\begin{aligned} 6x_1 - 2x_2 &= 14 \\ 9x_1 - x_2 + x_3 &= 21 \\ 3x_1 - 7x_2 + 5x_3 &= 9 \end{aligned}$$

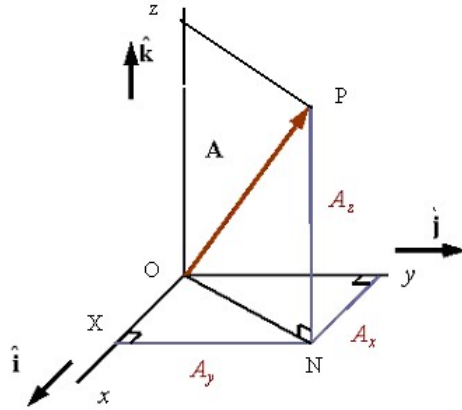


Figure 21

From figure 21

ΔONP ,

$$\begin{aligned}\vec{OP} &= \vec{A} = \vec{ON} + \vec{NP} \\ &= (\vec{OX} + \vec{XN}) + \vec{NP} \\ &= (\vec{OX} + \vec{OY}) + \vec{NP} \\ &= (\vec{OX} + \vec{OY}) + \vec{OZ} \\ &= A_x \hat{i} + A_y \hat{j} + A_z \hat{k}\end{aligned}$$

and

$$\begin{aligned}|\vec{A}| &= |\vec{OP}| = \sqrt{|\vec{OP}|^2} = \sqrt{|\vec{ON}|^2 + |\vec{NP}|^2} = \sqrt{|\vec{OX}|^2 + |\vec{XN}|^2 + |\vec{OZ}|^2} = \sqrt{A_x^2 + A_y^2 + A_z^2} \\ \vec{A} &= A_x \hat{i} + A_y \hat{j} + A_z \hat{k} \quad \text{and} \quad |\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}\end{aligned}$$

Q#02: Find the unit tangent (slope, ঢাল, স্পর্শক, $\tan \theta$, m , rate of change, derivative/differentiation) vector to the graph of $\vec{r}(t) = t^2 \hat{i} + t^3 \hat{j}$ at the point where $t = 2$

Answer # 02

$$\text{Given, } \vec{r}(t) = t^2 \hat{i} + t^3 \hat{j} \text{-----(i)}$$

$$\therefore \vec{r}(2) = 4 \hat{i} + 8 \hat{j} \text{-----(ii)}$$

Here $x = 4$, $y = 8$

$$\therefore \vec{OP} = \vec{r}(2) = 4 \hat{i} + 8 \hat{j} \quad [\text{See figure no 22}]$$

The tangent vector will be drawn at (4, 8) which is $\vec{T}(2)$

From (i),

$$\vec{r}(t) = t^2 \hat{i} + t^3 \hat{j}$$

$$\frac{d\vec{r}}{dt} = \vec{r}'(t) = 2t \hat{i} + 3t^2 \hat{j} \text{-----(iii)}$$

$$\therefore \vec{T}(t) = \vec{r}'(t) = 2t \hat{i} + 3t^2 \hat{j} \text{-----(iv)}$$

$$\therefore \vec{T}(2) = \vec{r}'(2) = 2 \times 2 \hat{i} + 3 \times 2^2 \hat{j}$$

$$\therefore \text{Tangent vector: } \vec{PQ} = \vec{T}(2) = \vec{r}'(2) = 4 \hat{i} + 12 \hat{j} \text{-----(v)}$$

$\therefore$ Unit tangent vector:

$$\begin{aligned} \frac{\vec{T}(2)}{|\vec{T}(2)|} &= \frac{\vec{r}'(2)}{|\vec{r}'(2)|} = \frac{4\hat{i} + 12\hat{j}}{\sqrt{4^2 + 12^2}} = \frac{4}{\sqrt{160}} \hat{i} + \frac{12}{\sqrt{160}} \hat{j} = \frac{4}{\sqrt{16 \times 10}} \hat{i} + \frac{12}{\sqrt{16 \times 10}} \hat{j} \\ &= \frac{4}{4\sqrt{10}} \hat{i} + \frac{12}{4\sqrt{10}} \hat{j} = \frac{1}{\sqrt{10}} \hat{i} + \frac{3}{\sqrt{10}} \hat{j} \end{aligned}$$

The Figure is following:

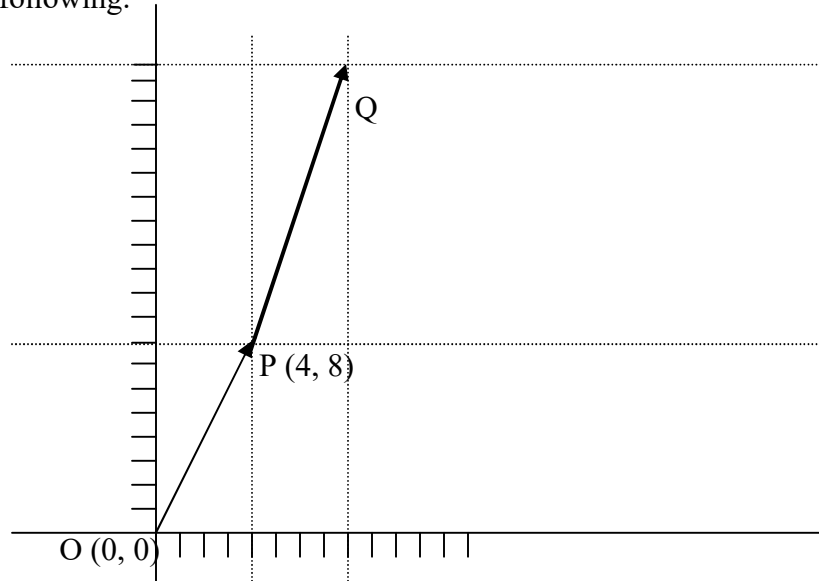


Figure: 22

When $t = 1$, $\vec{r}(1) = 1.\hat{i} + 1.\hat{j}$

When $t = 2$, $\vec{r}(2) = 4.\hat{i} + 8.\hat{j}$

When $t = 3$, $\vec{r}(3) = 9.\hat{i} + 27.\hat{j}$

From (iii),

$$\vec{r}'(t) = 2t \hat{i} + 3t^2 \hat{j}$$

When $t = 2$, $\vec{r}'(2) = 2 \times 2 \hat{i} + 3 \times 2^2 \hat{j} = 4 \hat{i} + 12 \hat{j}$ Answer

Vector Components

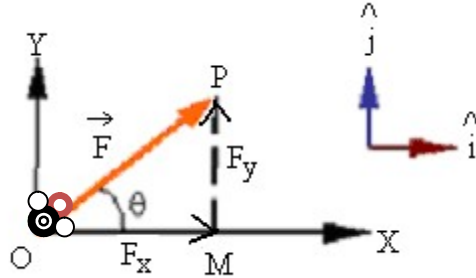


Figure 23

Here,

$$OM = F_x, MP = F_y, OP = |\vec{F}|$$

[$\because$ Any Vector = Length of this Vector $\times$ Unit Vector]

$$\vec{OM} = F_x \hat{i} \quad [\vec{OM} = F_x \hat{i} = \text{Length of this Vector} \times \text{Unit Vector}]$$

$$\vec{MP} = F_y \hat{j} \quad [\vec{MP} = F_y \hat{j} = \text{Length of this Vector} \times \text{Unit Vector}]$$

From, $\triangle OMP$,

$$\vec{OP} = \vec{F} = \vec{OM} + \vec{MP} = F_x \hat{i} + F_y \hat{j} \text{-----(i)}$$

From, $\triangle OMP$,

$$\frac{OM}{OP} = \cos \theta$$

$$OM = OP \cos \theta$$

$$F_x = |\vec{F}| \cos \theta \text{-----(ii)}$$

From, $\triangle OMP$,

$$\frac{MP}{OP} = \sin \theta$$

$$MP = OP \sin \theta$$

$$F_y = |\vec{F}| \sin \theta \text{-----(iii)}$$

The length of the vector (vector component) is: $\left| \vec{a} \right| = \frac{\vec{A} \cdot \vec{B}}{\left| \vec{B} \right|}$ -----(iv)

Again,

$$\vec{a} = \left| \vec{a} \right| \times \text{unit vector of } \vec{B}$$

$$\vec{a} = \left| \vec{a} \right| \times \frac{\vec{B}}{\left| \vec{B} \right|} \text{ -----(v)}$$

Putting the value of $\left| \vec{a} \right|$ from (iv) in (v)

$$\vec{a} = \left| \vec{a} \right| \times \frac{\vec{B}}{\left| \vec{B} \right|}$$

$$\vec{a} = \frac{\vec{A} \cdot \vec{B}}{\left| \vec{B} \right|^2} \times \frac{\vec{B}}{\left| \vec{B} \right|}$$

The projection of $\vec{A}$ on $\vec{B}$ is $\vec{a} = \frac{\vec{A} \cdot \vec{B}}{\left| \vec{B} \right|^2} \vec{B}$

----- (vi)

Q#03: Find the projection and components of the vector $\vec{A} = 2\hat{i} - \hat{j} + \hat{k}$ on the vector $\vec{B} = \hat{i} + 3\hat{j} - 2\hat{k}$

Answer: we have,

Vector component of $\vec{A}$ on $\vec{B}$ is $= \frac{\vec{A} \cdot \vec{B}}{\left| \vec{B} \right|}$

Given , $\vec{B} = \hat{i} + 3\hat{j} - 2\hat{k}$

$$\therefore \left| \vec{B} \right| = \sqrt{1^2 + 3^2 + (-2)^2} = \sqrt{1 + 9 + 4} = \sqrt{14}$$

$\therefore$ Vector component of $\vec{A}$ on $\vec{B}$ is

$$\begin{aligned}
 &= \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|} = \frac{(2\hat{i} - \hat{j} + \hat{k}) \cdot (\hat{i} + 3\hat{j} - 2\hat{k})}{\sqrt{14}} \\
 &= \frac{2 \cdot 1 - 1 \cdot 3 + 1 \cdot (-2)}{\sqrt{14}} \\
 &= \frac{2 - 3 - 2}{\sqrt{14}} \\
 &= \frac{-3}{\sqrt{14}}
 \end{aligned}$$

Again

$$\begin{aligned}
 \text{The projection of } \vec{A} \text{ on } \vec{B} \text{ is } \vec{a} &= \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|^2} \vec{B} \\
 &= \frac{-3}{(\sqrt{14})^2} (\hat{i} + 3\hat{j} - 2\hat{k}) \\
 &= \frac{-3}{14} (\hat{i} + 3\hat{j} - 2\hat{k}) \text{ Answer}
 \end{aligned}$$

Three –dimensional Coordinate System

The point P(a, b, c) determines a rectangular box as the figure

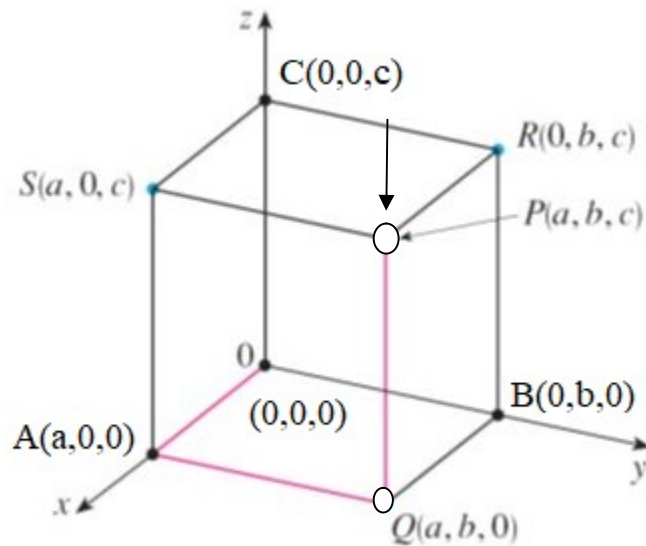


Figure: 27

Q#16: Show that $\vec{A} = \hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{B} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{C} = 3\hat{i} + \hat{j} - \hat{k}$ are coplanar.

Answer:

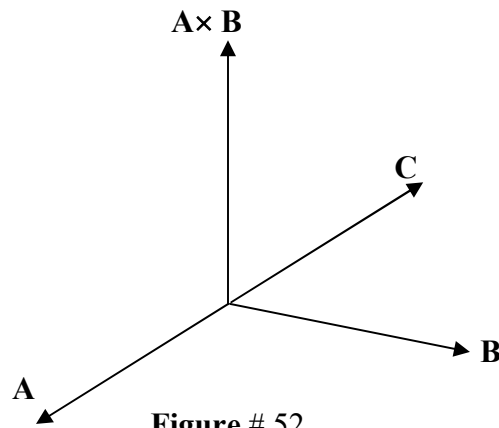


Figure # 52

If $\vec{A}$ is a third vector perpendicular to $(\vec{B} \times \vec{C})$, then $\vec{A}$, $\vec{B}$ and $\vec{C}$ are coplanar and $\vec{A} \cdot (\vec{B} \times \vec{C}) = 0$

Therefore, three vectors $\vec{A}$, $\vec{B}$, $\vec{C}$ are coplanar if $\vec{A} \cdot (\vec{B} \times \vec{C}) = 0$

$$\vec{B} \times \vec{C} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{vmatrix} = \hat{i}(1-2) - \hat{j}(-2-6) + \hat{k}(2+3) = -\hat{i} + 8\hat{j} + 5\hat{k}$$

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = (\hat{i} + 2\hat{j} - 3\hat{k}) \cdot (-\hat{i} + 8\hat{j} + 5\hat{k}) = -1 + 16 - 15 = 0$$

Therefore $\vec{A}$, $\vec{B}$, $\vec{C}$ are coplanar.

Q#17: On which condition of λ the vectors $\hat{i} + 2\hat{j} + 3\hat{k}$, $\lambda\hat{i} + 4\hat{j} + 7\hat{k}$, $-3\hat{i} - 2\hat{j} - 5\hat{k}$ are collinear

Solution

We know that if the points (x_1, y_1, z_1) , (x_2, y_2, z_2) and (x_3, y_3, z_3) be collinear then

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 & 2 & 3 \\ \lambda & 4 & 7 \\ -3 & -2 & -5 \end{vmatrix} = 0$$

$$\Rightarrow 1[4 \times (-5) - 7 \times (-2)] - 2[\lambda(-5) - (-3) \times 7] + 3[\lambda(-2) - 4(-3)] = 0$$

$$\Rightarrow 1[-20 + 14] - 2[-5\lambda + 21] + 3[-2\lambda + 12] = 0$$

$$\Rightarrow -6 + 10\lambda - 42 - 6\lambda + 36 = 0$$

$$\Rightarrow 4\lambda - 48 + 36 = 0$$

$$\Rightarrow 4\lambda - 12 = 0$$

$$\Rightarrow 4\lambda = 12$$

$$\Rightarrow \lambda = 3$$

Q#18: If $\vec{A} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{B} = \hat{i} + 3\hat{j} - 2\hat{k}$, $\vec{C} = -2\hat{i} + \hat{j} - 3\hat{k}$ and $\vec{D} = 3\hat{i} + 2\hat{j} + 5\hat{k}$, then find

the scalars **a, b, c** such that $\vec{D} = a\vec{A} + b\vec{B} + c\vec{C}$

Answer:

Given

$$\vec{D} = a\vec{A} + b\vec{B} + c\vec{C}$$

$$\Rightarrow 3\hat{i} + 2\hat{j} + 5\hat{k} = a(2\hat{i} - \hat{j} + \hat{k}) + b(\hat{i} + 3\hat{j} - 2\hat{k}) + c(-2\hat{i} + \hat{j} - 3\hat{k})$$

$$\Rightarrow 3\hat{i} + 2\hat{j} + 5\hat{k} = 2a\hat{i} - a\hat{j} + a\hat{k} + b\hat{i} + 3b\hat{j} - 2b\hat{k} - 2c\hat{i} + c\hat{j} - 3c\hat{k}$$

$$\Rightarrow 3\hat{i} + 2\hat{j} + 5\hat{k} = (2a + b - 2c)\hat{i} + (-a + 3b + c)\hat{j} + (a - 2b - 3c)\hat{k}$$

Equating the coefficient of $\hat{i}, \hat{j}, \hat{k}$

$$2a + b - 2c = 3 \text{ -----L}_1$$

$$-a + 3b + c = 2 \text{ -----L}_2$$

$$a - 2b - 3c = 5 \text{ -----L}_3$$

We have,

$$\mathbf{L}_i \rightarrow -\mathbf{a}_{i1}\mathbf{L}_1 + \mathbf{a}_{i1}\mathbf{L}_i$$

1st time:

$$\mathbf{i} = 2, \mathbf{L}_2 \rightarrow -\mathbf{a}_{21}\mathbf{L}_1 + \mathbf{a}_{11}\mathbf{L}_2$$

$$\rightarrow -(-1)(2a + b - 2c - 3) + 2(-a + 3b + c - 2)$$

$$\rightarrow 2a + b - 2c - 3 - 2a + 6b + 2c - 4$$

$$\rightarrow 7b - 7$$

$$\mathbf{i} = 3, \mathbf{L}_3 \rightarrow -\mathbf{a}_{31}\mathbf{L}_1 + \mathbf{a}_{11}\mathbf{L}_3$$

$$\rightarrow -1(2a + b - 2c - 3) + 2(a - 2b - 3c - 5)$$

$$\rightarrow -2a - b + 2c + 3 + 2a - 4b - 6c - 10$$

$$\rightarrow -5b - 4c - 7$$

Thus we obtain the following new system

$$2a + b - 2c = 3$$

$$7b = 7$$

$$-5b - 4c = 7$$

Thus we obtain the following new system

$$2a + b - 2c = 3$$

$$-5b - 4c = 7$$

$$7b = 7$$

Thus we obtain the following new system

$$2a + b - 2c = 3$$

$$-5b - 4c = 7$$

$$\therefore b = 1$$

$$\text{Then } -5b - 4c = 7$$

$$-5(1) - 4c = 7$$

$$-4c = 7 + 5$$

$$-4c = 12$$

$$\therefore c = -3$$

$$\text{And } 2a + b - 2c = 3$$

$$2a + 1 - 2(-3) = 3$$

$$2a + 1 + 6 = 3$$

$$2a + 7 = 3$$

$$2a = 3 - 7$$

$$2a = -4$$

$$a = -2$$

Hence $(a, b, c) = (-2, 1, -3)$ Answer

Q# 19: Figure no 53 shows a force $\vec{F}$ of 100 N applied in the positive z-direction at the point $Q(1,1,1)$ of a cube whose sides have a length of 1 m. assuming that the cube is free to rotate about the point $P(0,0,0)$ (the origin), find the scalar moment of the force about P and describe the direction of rotation.

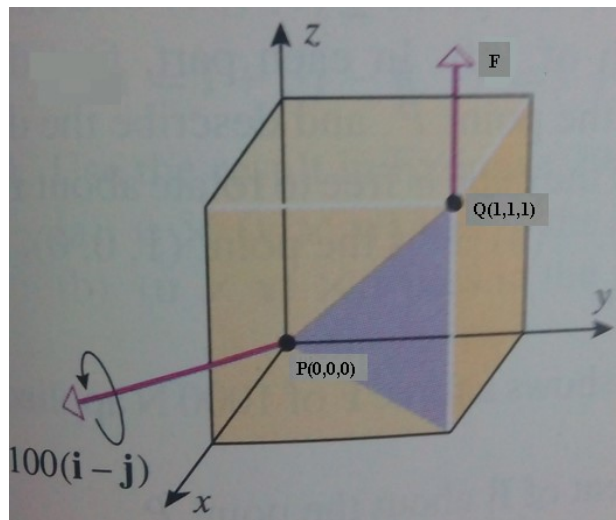


Figure 53

$$\vec{V} = \frac{d\vec{r}}{dt} = \frac{d}{dt}[2t^2 \hat{i}] + \frac{d}{dt}[(t^2 - 4t) \hat{j}] + \frac{d}{dt}[(3t - 5) \hat{k}]$$

$$\vec{V} = \frac{d\vec{r}}{dt} = 4t \hat{i} + (2t - 4) \hat{j} + 3 \hat{k} \text{-----(i)}$$

The Velocity at $t = 1$;

$$\vec{V} = \frac{d\vec{r}}{dt} = 4t \hat{i} + (2t - 4) \hat{j} + 3 \hat{k}$$

$$\vec{V} = \frac{d\vec{r}}{dt} = 4.1 \hat{i} + (2.1 - 4) \hat{j} + 3 \hat{k}$$

$$\vec{V} = \frac{d\vec{r}}{dt} = 4 \hat{i} - 2 \hat{j} + 3 \hat{k} \text{-----(ii)}$$

The unit vector of $\vec{a}$ is $\hat{e} = \frac{\vec{a}}{|\vec{a}|} = \frac{\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{1^2 + (-3)^2 + 2^2}} = \frac{\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{1+9+4}} = \frac{\hat{i} - 3\hat{j} + 2\hat{k}}{\sqrt{14}}$

$$= \frac{\hat{i}}{\sqrt{14}} - \frac{3\hat{j}}{\sqrt{14}} + \frac{2\hat{k}}{\sqrt{14}} \text{-----(iii)}$$

The component of the velocity in the given direction $\vec{a} = \hat{i} - 3\hat{j} + 2\hat{k}$ is $\vec{V} \cdot \hat{e}$, where $\hat{e}$ is a unit vector in the direction of $\vec{a}$.

$$\vec{V} \cdot \hat{e} = (4\hat{i} - 2\hat{j} + 3\hat{k}) \cdot \left(\frac{\hat{i}}{\sqrt{14}} - \frac{3\hat{j}}{\sqrt{14}} + \frac{2\hat{k}}{\sqrt{14}} \right)$$

$$\vec{V} \cdot \hat{e} = \left(\frac{4}{\sqrt{14}} + \frac{6}{\sqrt{14}} + \frac{6}{\sqrt{14}} \right)$$

$$\vec{V} \cdot \hat{e} = \frac{16}{\sqrt{14}} \text{ Answer}$$

Q#21: A particle moves so that its position vector is given by $\vec{r} = \cos \omega t \hat{i} + \sin \omega t \hat{j}$, where ω is a constant. Show that (a) the velocity $\vec{V}$ of the particle is perpendicular to $\vec{r}$, (b) The acceleration $\vec{a}$ is directed toward the origin and has magnitude proportional to the distance from the origin (c) $\vec{r} \times \vec{V}$ = a constant vector

Answer: Given, $\vec{r} = \cos \omega t \hat{i} + \sin \omega t \hat{j}$

a) Then the velocity is $\vec{V} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(\cos \omega t) \hat{i} + \frac{d}{dt}(\sin \omega t) \hat{j}$

$$\vec{V} = \frac{d\vec{r}}{dt} = -\omega \sin \omega t \hat{i} + \omega \cos \omega t \hat{j}$$

$O_p = r$
 $O_p = -r$

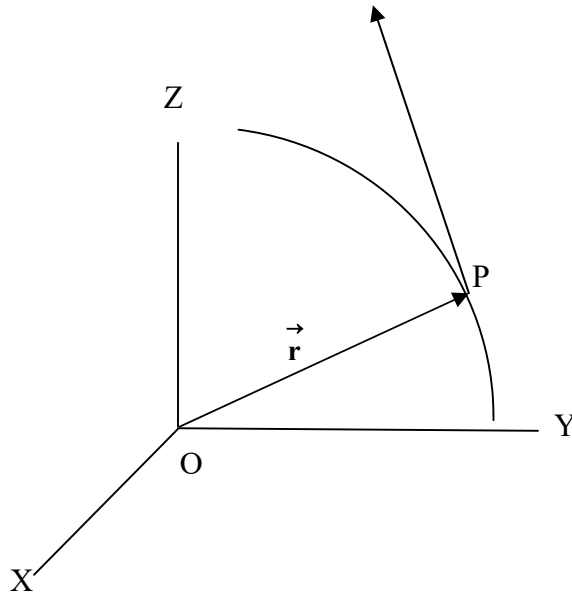


Figure # 59

Then $\vec{r} \cdot \vec{V} = (\cos \omega t \hat{i} + \sin \omega t \hat{j}) \cdot (-\omega \sin \omega t \hat{i} + \omega \cos \omega t \hat{j})$

Then $\vec{r} \cdot \vec{V} = (\cos \omega t)(-\omega \sin \omega t) + (\sin \omega t)(\omega \cos \omega t)$

Then $\vec{r} \cdot \vec{V} = -\omega \sin \omega t \cos \omega t + \omega \sin \omega t \cos \omega t$

Then $\vec{r} \cdot \vec{V} = 0$

Hence $\vec{r}$ and $\vec{V}$ are perpendicular.

b) The acceleration is: $\vec{a} = \frac{d\vec{V}}{dt} = \frac{d}{dt} (-\omega \sin \omega t \hat{i} + \omega \cos \omega t \hat{j})$

The acceleration is: $\vec{a} = \frac{d\vec{V}}{dt} = -\omega^2 \cos \omega t \hat{i} - \omega^2 \sin \omega t \hat{j}$

The acceleration is: $\vec{a} = \frac{d\vec{V}}{dt} = -\omega^2 (\cos \omega t \hat{i} + \sin \omega t \hat{j}) = -\omega^2 \vec{r}$

$$\vec{a} = -\omega^2 \vec{r} \text{-----(i)}$$

$$\vec{a} \propto \vec{r}$$

$$\left| \vec{a} \right| \propto \left| \vec{r} \right| \text{-----(ii)}$$

From equation (1), the acceleration is opposite to the direction of $\vec{r}$. i.e. it is directed toward the origin, Its magnitude is proportional to $\left| \vec{r} \right|$ which is the distance from the origin.

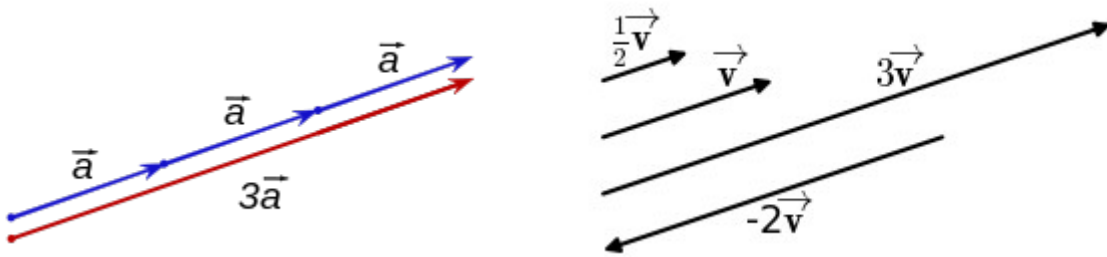


Figure # 60

c) Here $\vec{r} = \cos \omega t \hat{i} + \sin \omega t \hat{j}$ and $\vec{V} = -\omega \sin \omega t \hat{i} + \omega \cos \omega t \hat{j}$

$$\begin{aligned} \vec{r} \times \vec{V} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos \omega t & \sin \omega t & 0 \\ -\omega \sin \omega t & \omega \cos \omega t & 0 \end{vmatrix} \\ &= \hat{i}(\sin \omega t \times 0 - 0 \times \omega \cos \omega t) - \hat{j}(\cos \omega t \times 0 - (-\omega \sin \omega t) \times 0) + \hat{k}(\cos \omega t \times \omega \cos \omega t - (-\omega \sin \omega t) \times \sin \omega t) \\ &= \hat{i} \times 0 - \hat{j} \times 0 + \hat{k}(\omega \cos^2 \omega t + \omega \sin^2 \omega t) \\ &= \hat{k} \omega (\cos^2 \omega t + \sin^2 \omega t) \\ &= \hat{k} \omega.1 \quad [\because \cos^2 \omega t + \sin^2 \omega t = 1] \\ &= \hat{k} \omega \end{aligned}$$

Q#22: A particle moves along a circular path in such a way that its x- and y-coordinates at time t are $x = 2 \cos t$, $y = 2 \sin t$

a) Find the instantaneous velocity and speed of the particle at time t.

$$\vec{V}(t) \cdot \vec{a}(t) = (-2 \sin t \hat{i} + 2 \cos t \hat{j}) \cdot (-2 \cos t \hat{i} - 2 \sin t \hat{j}) = 4 \sin t \cos t - 4 \sin t \cos t = 0$$

Since the dot product of Velocity Vector (ii) and acceleration Vector (v) is Zero, Hence acceleration vector is perpendicular to the velocity vector.

(Proved)

✓ **Q#23:** A particle moves through 3-space in such a way that its velocity is $\vec{v}(t) = \hat{i} + t \hat{j} + t^2 \hat{k}$. Find the co-ordinates of the particle at time $t = 1$ given that the particle is at the point $(-1, 2, 4)$ at time $t = 0$

Answer: We have,

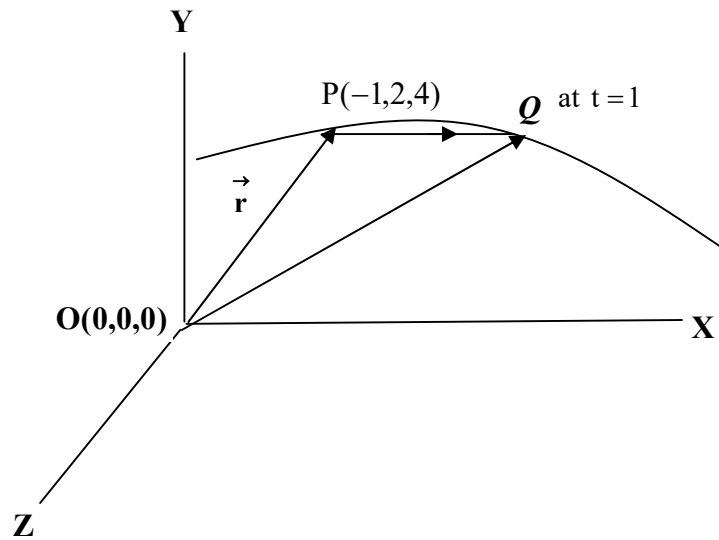


Figure 62

$$\vec{v}(t) = \frac{d\vec{r}}{dt} \text{ Where } \vec{r} \text{ is a position vector.}$$

$$\text{Given, } \vec{v}(t) = \hat{i} + t \hat{j} + t^2 \hat{k}$$

$$\therefore \vec{v}(t) = \frac{d\vec{r}(t)}{dt} = \hat{i} + t \hat{j} + t^2 \hat{k}$$

$$\frac{d\vec{r}(t)}{dt} = \hat{i} + t \hat{j} + t^2 \hat{k}$$

$$d\vec{r}(t) = (\hat{i} + t \hat{j} + t^2 \hat{k}) dt \text{ -----(i)}$$

Integrate (i) both sides, we get,

$$\int d\vec{r}(t) = \int (\hat{i} + t \hat{j} + t^2 \hat{k}) dt$$

$$\int d\vec{r}(t) = \int \hat{i} dt + \int t \hat{j} dt + \int t^2 \hat{k} dt$$

$$\vec{r}(t) = t \hat{i} + \frac{t^2}{2} \hat{j} + \frac{t^3}{3} \hat{k} + C \quad \text{-----(ii)}$$

Where C is a vector constant of integration. Since the coordinates of the particle at time $t = 0$ are $(-1, 2, 4)$, the position vector at time $t = 0$ is

We have the position vector

$$\vec{r}(t) = x \hat{i} + y \hat{j} + z \hat{k} \quad [\text{Page no 48, Figure no 57, Equation no (i)}]$$

$$\vec{r}(0) = (-1) \hat{i} + 2 \hat{j} + 4 \hat{k} \quad [\text{at time } t = 0, \text{ the position vector is at } (-1, 2, 4)]$$

$$\vec{r}(0) = (-1) \hat{i} + 2 \hat{j} + 4 \hat{k} \quad \text{-----(iii)}$$

Again, putting $t = 0$ in (ii), we get,

$$\vec{r}(t) = t \hat{i} + \frac{t^2}{2} \hat{j} + \frac{t^3}{3} \hat{k} + C$$

$$\vec{r}(0) = 0 \hat{i} + \frac{0^2}{2} \hat{j} + \frac{0^3}{3} \hat{k} + C$$

$$\vec{r}(0) = 0 + 0 + 0 + C$$

$$\vec{r}(0) = C \quad \text{-----(iv)}$$

Comparing (iii) and (iv), we get,

$$\vec{r}(0) = (-1) \hat{i} + 2 \hat{j} + 4 \hat{k} = C$$

$$C = -\hat{i} + 2 \hat{j} + 4 \hat{k} \quad \text{-----(v)}$$

Putting the value of C in (ii), we get,

$$\vec{r}(t) = t \hat{i} + \frac{t^2}{2} \hat{j} + \frac{t^3}{3} \hat{k} + C$$

$$\vec{r}(t) = t \hat{i} + \frac{t^2}{2} \hat{j} + \frac{t^3}{3} \hat{k} - \hat{i} + 2 \hat{j} + 4 \hat{k}$$

$$\vec{r}(t) = (t-1) \hat{i} + \left(\frac{t^2}{2} + 2\right) \hat{j} + \left(\frac{t^3}{3} + 4\right) \hat{k} \quad \text{-----(vi)}$$

Thus, at time $t = 1$, the position vector of the particle is

From (vi),

$$\vec{r}(t) = (t-1) \hat{i} + \left(\frac{t^2}{2} + 2\right) \hat{j} + \left(\frac{t^3}{3} + 4\right) \hat{k}$$

$$\vec{r}(1) = (1-1) \hat{i} + \left(\frac{1^2}{2} + 2\right) \hat{j} + \left(\frac{1^3}{3} + 4\right) \hat{k}$$

$$\vec{r}(1) = 0 \hat{i} + \frac{5}{2} \hat{j} + \frac{13}{3} \hat{k}.$$

So, the coordinates of the particle at time $t = 1$ is $(0, \frac{5}{2}, \frac{13}{3})$ the Answer

$$\therefore \frac{d\phi}{dx} = 2xy + x^2 \frac{dy}{dx}.$$

Again

$$\phi(x, y) = x^2 y$$

$$\therefore \frac{d\phi}{dy} = \frac{d}{dy}(x^2 y)$$

$$= x^2 \frac{d}{dy}(y) + y \frac{d}{dy}(x^2) \left[\because \frac{d}{dx}(uv) = u \frac{d}{dx}(v) + v \frac{d}{dx}(u) \right]$$

$$= x^2 \times 1 + y \times 2x \frac{dx}{dy}$$

$$\therefore \frac{d\phi}{dy} = x^2 + 2xy \frac{dx}{dy}$$

$$\therefore \frac{d\phi}{dy} = 2xy \frac{dx}{dy} + x^2$$

We know,

$$d\phi = \frac{\partial \phi}{\partial x} \times dx + \frac{\partial \phi}{\partial y} \times dy \quad \text{-----(iii)}$$

$$\Rightarrow d\phi = 2xy \times dx + x^2 \times dy \quad \text{-----(iv) [From (i) & (ii)]}$$

$$\Rightarrow \frac{d\phi}{dx} = 2xy \times \frac{dx}{dx} + x^2 \frac{dy}{dx} \quad \text{[Dividing both sides by dx]}$$

$$\therefore \frac{d\phi}{dx} = 2xy + x^2 \frac{dy}{dx}.$$

From equation (iii),

$$d\phi = \frac{\partial \phi}{\partial x} \times dx + \frac{\partial \phi}{\partial y} \times dy$$

$$\Rightarrow d\phi = 2xy \times dx + x^2 \times dy$$

$$\Rightarrow \frac{d\phi}{dy} = 2xy \frac{dx}{dy} + x^2 \frac{dy}{dy} \quad \text{[Dividing by dy]}$$

$$\therefore \frac{d\phi}{dy} = 2xy \frac{dx}{dy} + x^2$$

Q#31: Show that $\vec{\nabla} \phi$ is a vector perpendicular to the surface $\phi(x, y, z) = c$, where c is a constant.

Answer: Let $\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$ be the position vector to any point $P(x, y, z)$ on the surface. [Page no 48, Figure no 57, Equation no (i)]

$\therefore d\vec{r} = dx \hat{i} + dy \hat{j} + dz \hat{k}$ lies in the tangent plane to the surface at P.

Given, $\phi(x, y, z) = c$

$$\Rightarrow d\phi = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz = d(c)$$

$$\Rightarrow d\phi = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz = 0$$

$$\Rightarrow \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz = 0 \text{-----(i)}$$

$$\Rightarrow \left(\frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}\right) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = 0 \text{-----(ii)}$$

$$\Rightarrow \left(\frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}\right) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = 0$$

$$\Rightarrow \vec{\nabla}\phi \cdot d\vec{r} = 0 \quad \left[\because \vec{\nabla} = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} ; d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k} \right]$$

So that $\vec{\nabla}\phi$ perpendicular to $d\vec{r}$ and therefore to the surface.

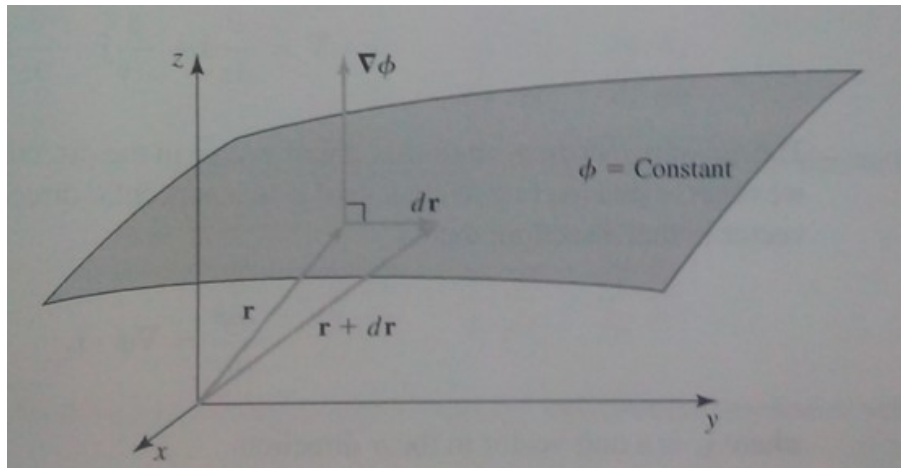


Figure # 72

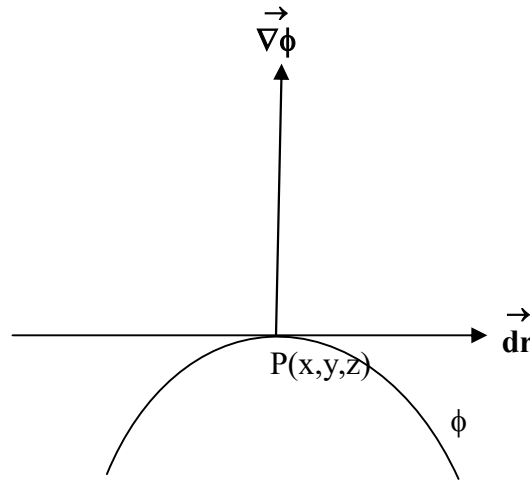


Figure # 73

It is clear that the vector $\vec{\nabla} \phi$ is perpendicular (normal) to the tangent vector $\vec{dr}$ at a point $P(x, y, z)$ that is $\vec{\nabla} \phi \cdot d\vec{r} = 0$

So we conclude that $\vec{\nabla} \phi$ is normal (perpendicular) vector to the surface $\phi(x, y, z) = c$ at (x, y, z) .

[N.B. We always remember that $\vec{\nabla} \phi$ is perpendicular to the tangent to the surface but not with surface directly and $\vec{\nabla} \phi$ is normal to the surface $\phi(x, y, z) = c$, Where $\vec{\nabla} \phi$ is a vector component that is $\vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi$

Q# 32: Find a unit normal to the surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$

Answer: Given, $\phi(x, y, z) = x^2y + 2xz = 4$

$$\begin{aligned} \therefore \vec{\nabla} \phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi = \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) = \hat{i} \frac{\partial}{\partial x} \phi + \hat{j} \frac{\partial}{\partial y} \phi + \hat{k} \frac{\partial}{\partial z} \phi \\ &= \hat{i} \frac{\partial}{\partial x} (x^2y + 2xz) + \hat{j} \frac{\partial}{\partial y} (x^2y + 2xz) + \hat{k} \frac{\partial}{\partial z} (x^2y + 2xz) \\ &= (2xy + 2z) \hat{i} + x^2 \hat{j} + 2x \hat{k} \\ &= (2 \times 2 \times (-2) + 2 \times 3) \hat{i} + 2^2 \hat{j} + 2 \times 2 \hat{k} \text{ at the point } (2, -2, 3) \\ &= -2 \hat{i} + 4 \hat{j} + 4 \hat{k} \end{aligned}$$

$$\text{Then a unit normal to the surface} = \frac{-2 \hat{i} + 4 \hat{j} + 4 \hat{k}}{\sqrt{(-2)^2 + (4)^2 + (4)^2}} = -\frac{1}{3} \hat{i} + \frac{2}{3} \hat{j} + \frac{2}{3} \hat{k} \text{ Answer}$$

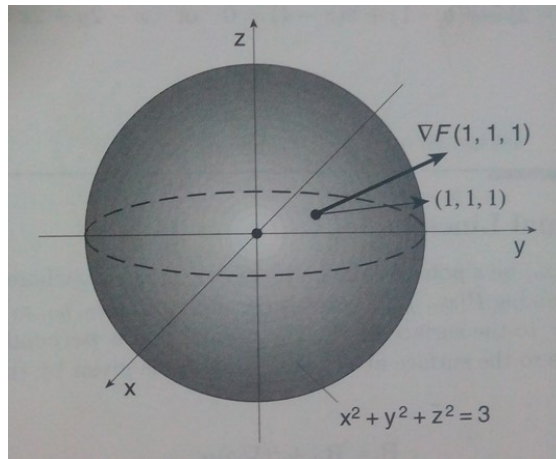


Figure # 74

Q#34: Prove that the angle between the surfaces at the point is equal to the angle between the normals to the surfaces at the point.

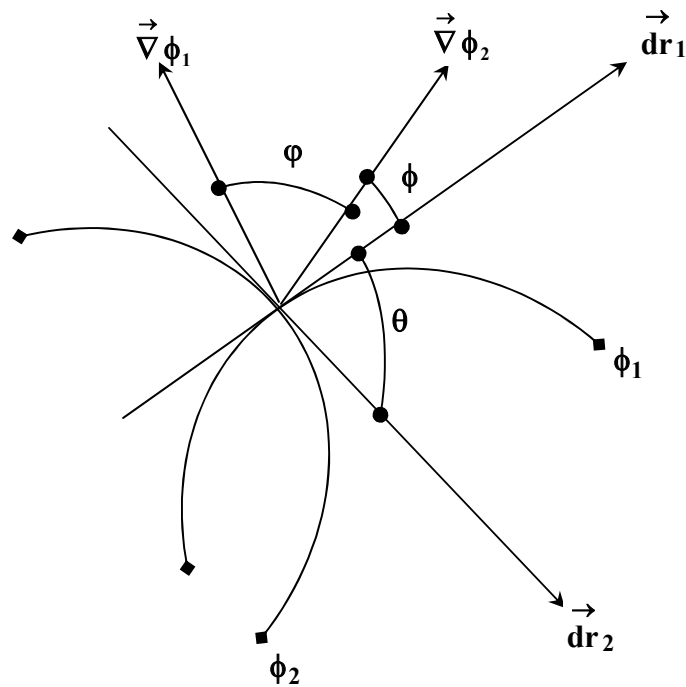


Figure # 75

Here, $\hat{\eta}_1$ is the unit vector of $\vec{\nabla} \phi_1$ and $\hat{\eta}_2$ is the unit vector of $\vec{\nabla} \phi_2$

We can write,

$$\hat{\eta}_1 = \frac{\vec{\nabla} \phi_1}{|\vec{\nabla} \phi_1|} \text{ and } \hat{\eta}_2 = \frac{\vec{\nabla} \phi_2}{|\vec{\nabla} \phi_2|}$$

We have,

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$\therefore \hat{\eta}_1 \cdot \hat{\eta}_2 = |\hat{\eta}_1| |\hat{\eta}_2| \cos \varphi \quad [\because \varphi \text{ be the angle between the normals to the surfaces } \phi_1 \text{ and } \phi_2]$$

$$\therefore \hat{\eta}_1 \cdot \hat{\eta}_2 = 1.1 \cos \varphi \quad [\because \text{The length or magnitude of unit vector is 1}]$$

$$\Rightarrow \frac{\vec{\nabla} \phi_1}{|\vec{\nabla} \phi_1|} \cdot \frac{\vec{\nabla} \phi_2}{|\vec{\nabla} \phi_2|} = 1.1 \cos \varphi$$

$$\Rightarrow \vec{\nabla} \phi_1 \cdot \vec{\nabla} \phi_2 = |\vec{\nabla} \phi_1| |\vec{\nabla} \phi_2| \cos \varphi \text{-----(i)}$$

Again, from figure # 75

$$\varphi + \phi = 90$$

$$\pm \theta \pm \phi = \pm 90$$

$$\varphi - \theta = 0$$

$$\therefore \varphi = \theta \text{-----(ii) (Proved)}$$

Q# 35: Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$

Answer:

$$\text{Given, } z = x^2 + y^2 - 3$$

$$\Rightarrow x^2 + y^2 - z = 3$$

$$\text{Let, } \phi_1(x, y, z) = x^2 + y^2 + z^2 = 9 \text{ and } \phi_2(x, y, z) = x^2 + y^2 - z = 3$$

$$\therefore \nabla \phi_1 = \left(\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \phi_1 = \left(\frac{\delta \phi_1}{\delta x} \hat{i} + \frac{\delta \phi_1}{\delta y} \hat{j} + \frac{\delta \phi_1}{\delta z} \hat{k} \right) = \hat{i} \frac{\delta}{\delta x} \phi_1 + \hat{j} \frac{\delta}{\delta y} \phi_1 + \hat{k} \frac{\delta}{\delta z} \phi_1$$

$$= \hat{i} \frac{\delta}{\delta x} (x^2 + y^2 + z^2) + \hat{j} \frac{\delta}{\delta y} (x^2 + y^2 + z^2) + \hat{k} \frac{\delta}{\delta z} (x^2 + y^2 + z^2)$$

$$= 2x \hat{i} + 2y \hat{j} + 2z \hat{k}$$

$$= 4 \hat{i} - 2 \hat{j} + 4 \hat{k} \text{ at the point } (2, -1, 2)$$

A normal to $x^2 + y^2 + z^2 = 9$ at $(2, -1, 2)$ is $\nabla\phi_1 = 4\hat{i} - 2\hat{j} + 4\hat{k}$

Again,

$$\phi_2(x, y, z) = x^2 + y^2 - z = 3$$

$$\begin{aligned}\therefore \nabla\phi_2 &= \left(\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}\right)\phi_2 = \left(\frac{\delta\phi_2}{\delta x}\hat{i} + \frac{\delta\phi_2}{\delta y}\hat{j} + \frac{\delta\phi_2}{\delta z}\hat{k}\right) = \hat{i}\frac{\delta}{\delta x}\phi_2 + \hat{j}\frac{\delta}{\delta y}\phi_2 + \hat{k}\frac{\delta}{\delta z}\phi_2 \\ &= \hat{i}\frac{\delta}{\delta x}(x^2 + y^2 - z) + \hat{j}\frac{\delta}{\delta y}(x^2 + y^2 - z) + \hat{k}\frac{\delta}{\delta z}(x^2 + y^2 - z) \\ &= 2x\hat{i} + 2y\hat{j} - \hat{k} \\ &= 4\hat{i} - 2\hat{j} - \hat{k} \text{ at the point } (2, -1, 2)\end{aligned}$$

A normal to $x^2 + y^2 - z = 3$ at $(2, -1, 2)$ is $\nabla\phi_2 = 4\hat{i} - 2\hat{j} - \hat{k}$

Let ϕ be the angle between the normals to the surfaces at the point $(2, -1, 2)$

Then we have,

$$\begin{aligned}\nabla\phi_1 \cdot \nabla\phi_2 &= |\nabla\phi_1| |\nabla\phi_2| \cos\phi, & [\text{from equation no (i), page no 78}] \\ \Rightarrow (4\hat{i} - 2\hat{j} + 4\hat{k}) \cdot (4\hat{i} - 2\hat{j} - \hat{k}) &= \left|4\hat{i} - 2\hat{j} + 4\hat{k}\right| \left|4\hat{i} - 2\hat{j} - \hat{k}\right| \cos\phi \\ \Rightarrow 16 + 4 - 4 &= \sqrt{(4)^2 + (-2)^2 + (4)^2} \sqrt{(4)^2 + (-2)^2 + (-1)^2} \cos\phi \\ \Rightarrow 16 &= 6\sqrt{21} \cos\phi \\ \Rightarrow \cos\phi &= \frac{16}{6\sqrt{21}} \\ \therefore \phi &= \cos^{-1}\left(\frac{16}{6\sqrt{21}}\right) \text{ Answer } [\because \phi = \theta]\end{aligned}$$

Q# 36: Prove that $\nabla \cdot \left[\frac{\vec{f(r)}}{r} \right] = \frac{1}{r^2} \frac{d}{dr} [r^2 f(r)]$

Here $\left| \vec{r} \right| = r$

L.H.S.

$$\begin{aligned}&\nabla \cdot \left[\frac{\vec{f(r)}}{r} \right] \\ &= \left(\hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z} \right) \cdot \left[\frac{\vec{f(r)}}{r} (x\hat{i} + y\hat{j} + z\hat{k}) \right] [\because \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}] \\ &= \left(\hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z} \right) \cdot \left(\hat{i} \frac{f(r)}{r} x + \hat{j} \frac{f(r)}{r} y + \hat{k} \frac{f(r)}{r} z \right) \\ &= \frac{\delta f(r)}{\delta x} \frac{x}{r} + \frac{\delta f(r)}{\delta y} \frac{y}{r} + \frac{\delta f(r)}{\delta z} \frac{z}{r} \text{------(i)} \\ &[\because \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1; \hat{k} \cdot \hat{k} = 1]\end{aligned}$$

direction of $\hat{a}$ and $\frac{d\phi}{ds} = \hat{a} \cdot \text{grad } \phi$. Grad ϕ will be a maximum when $\hat{a}$ and grad ϕ have the same direction, since then, $\hat{a} \cdot \text{grad } \phi = \|\hat{a}\| \|\text{grad } \phi\| \cos \theta$ and θ will be zero

Thus direction of grad ϕ gives the direction in which the maximum rate of change of ϕ occurs.

Q# 37: Find the directional derivative of the function $\phi = x^2z + 2xy^2 + yz^2$ at the point of (1, 2, -1) in the direction of the vector $\vec{A} = 2\hat{i} + 3\hat{j} + \hat{k}$.

We start off with $\phi = x^2z + 2xy^2 + yz^2$

$$\therefore \nabla \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi$$

$$\therefore \nabla \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2z + 2xy^2 + yz^2)$$

$$\therefore \nabla \phi =$$

$$\hat{i} \frac{\partial}{\partial x} (x^2z + 2xy^2 + yz^2) + \hat{j} \frac{\partial}{\partial y} (x^2z + 2xy^2 + yz^2) + \hat{k} \frac{\partial}{\partial z} (x^2z + 2xy^2 + yz^2)$$

$$\therefore \nabla \phi = \hat{i}(2xz + 2.1.y^2 + 0) + \hat{j}(0 + 2x.2y + 1.z^2) + \hat{k}(x^2.1 + 0 + y.2z)$$

$$\therefore \nabla \phi = \hat{i}(2xz + 2y^2) + \hat{j}(4xy + z^2) + \hat{k}(x^2 + 2yz)$$

At the point (1, 2, -1)

$$\nabla \phi = \hat{i}(2xz + 2y^2) + \hat{j}(4xy + z^2) + \hat{k}(x^2 + 2yz)$$

$$\therefore \nabla \phi = \hat{i}[2.1.(-1) + 2(2^2)] + \hat{j}[4.1.2 + (-1)^2] + \hat{k}[(1^2 + 2.2.(-1))]$$

$$\therefore \nabla \phi = \hat{i}[-2 + 8] + \hat{j}[8 + 1] + \hat{k}[(1 - 4)]$$

$$\therefore \nabla \phi = \hat{i}[6] + \hat{j}[9] + \hat{k}(-3)$$

$$\therefore \nabla \phi = 6\hat{i} + 9\hat{j} - 3\hat{k}$$

Next we have to find out the unit vector $\hat{a}$ where $\vec{A} = 2\hat{i} + 3\hat{j} - 4\hat{k}$

$$\therefore \hat{a} = \frac{\vec{A}}{|\vec{A}|}$$

$$\therefore \hat{a} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{\sqrt{2^2 + 3^2 + (-4)^2}}$$

$$\therefore \hat{a} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{|4 + 9 + 16|}$$

$$\therefore \hat{a} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{|29|}$$

$$\text{Hence } \frac{d\phi}{ds} = \hat{a} \cdot \vec{\nabla} \phi$$

$$\frac{d\phi}{ds} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{|29|} \cdot (6\hat{i} + 9\hat{j} - 3\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{|29|} (2\hat{i} + 3\hat{j} - 4\hat{k}) \cdot (6\hat{i} + 9\hat{j} - 3\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{|29|} (12 + 27 + 12)$$

$$\frac{d\phi}{ds} = \frac{51}{|29|} \text{ Answer}$$

Q#38: Find the directional derivative of the function $\phi = x^2y + y^2z + z^2x$ at the point of (1, -1, 2) in the direction of the vector $\vec{A} = 4\hat{i} + 2\hat{j} - 5\hat{k}$.

We have $\phi = x^2y + y^2z + z^2x$

$$\therefore \nabla \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi$$

$$\therefore \nabla \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2y + y^2z + z^2x)$$

$$\therefore \nabla \phi = \hat{i} \frac{\partial}{\partial x} (x^2y + y^2z + z^2x) + \hat{j} \frac{\partial}{\partial y} (x^2y + y^2z + z^2x) + \hat{k} \frac{\partial}{\partial z} (x^2y + y^2z + z^2x)$$

$$\therefore \nabla \phi = \hat{i}(2xy + 0 + z^2 \cdot 1) + \hat{j}(x^2 \cdot 1 + 2yz + 0) + \hat{k}(0 + y^2 \cdot 1 + 2zx)$$

$$\therefore \nabla \phi = \hat{i}(2xy + z^2) + \hat{j}(x^2 + 2yz) + \hat{k}(y^2 + 2zx)$$

At the point (1, -1, 2)

$$\therefore \nabla \phi = \hat{i}(2xy + z^2) + \hat{j}(x^2 + 2yz) + \hat{k}(y^2 + 2zx)$$

$$\therefore \nabla \phi = \hat{i}[2 \cdot 1 \cdot (-1) + 2^2] + \hat{j}[1^2 + 2(-1) \cdot 2] + \hat{k}[(-1)^2 + 2 \cdot 2 \cdot 1]$$

$$\therefore \nabla \phi = \hat{i}[-2 + 4] + \hat{j}[1 - 4] + \hat{k}[1 + 4]$$

$$\therefore \nabla \phi = 2\hat{i} - 3\hat{j} + 5\hat{k}$$

Next we have to find out the unit vector $\hat{a}$ where $\vec{A} = 4\hat{i} + 2\hat{j} - 5\hat{k}$

$$\frac{\partial \phi}{\partial z} = \frac{\partial}{\partial z}(x^2 - y^2 + 2z^2)$$

$$\frac{\partial \phi}{\partial z} = (0 - 0 + 4z)$$

$$\frac{\partial \phi}{\partial z} = 4z$$

$$\phi(x, y, z) = x^2 - y^2 + 2z^2$$

$$\text{grad } \phi = \vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi$$

$$\text{grad } \phi = \vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2 - y^2 + 2z^2)$$

$$\text{grad } \phi = \vec{\nabla} \phi = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x^2 - y^2 + 2z^2)$$

$$\text{grad } \phi = \vec{\nabla} \phi = \hat{i} \frac{\partial}{\partial x} (x^2 - y^2 + 2z^2) + \hat{j} \frac{\partial}{\partial y} (x^2 - y^2 + 2z^2) + \hat{k} \frac{\partial}{\partial z} (x^2 - y^2 + 2z^2)$$

$$\text{grad } \phi = \vec{\nabla} \phi = \hat{i} 2x + \hat{j} (-2y) + \hat{k} 4z$$

$$\text{grad } \phi = \vec{\nabla} \phi(1, 2, 3) = \hat{i} (2 \times 1) + \hat{j} (-2 \times 2) + \hat{k} (4 \times 3)$$

$$\text{grad } \phi = \vec{\nabla} \phi(1, 2, 3) = 2\hat{i} - 4\hat{j} + 12\hat{k}$$

Given ,

$$\vec{A} = 4\hat{i} - 2\hat{j} + \hat{k}$$

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|} = \frac{4\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{(4)^2 + (-2)^2 + (1)^2}} = \frac{4\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{16 + 4 + 1}} = \frac{4\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{21}}$$

∴ The directional derivative of the function $\phi = (x, y, z) = x^2 - y^2 + 2z^2$ at the point of

$$(1, 2, 3) \text{ in the direction of the vector } \vec{A} = \vec{\nabla} \phi. \hat{A} = (2\hat{i} - 4\hat{j} + 12\hat{k}). \frac{4\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{21}}$$

$$= \frac{8 + 8 + 12}{\sqrt{21}} = \frac{28}{\sqrt{21}}$$

Q# 40 Suppose that over a certain region of space the electrical potential V is given by

$$\phi(x, y, z) = 5x^2 - 3xy + xyz$$

- (i) Find the rate of change (derivative) of the potential at P(3, 4, 3) in the direction of the vector $\vec{v} = \hat{i} + \hat{j} - \hat{k}$
- (ii) In which direction does ϕ changes most rapidly at P?

(iii) What is the maximum rate of change at P?

$$\begin{aligned}
 \text{Answer: grad } \phi &= \vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi \\
 &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (5x^2 - 3xy + xyz) \\
 &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (5x^2 - 3xy + xyz) \\
 &= \hat{i} \frac{\partial}{\partial x} (5x^2 - 3xy + xyz) + \hat{j} \frac{\partial}{\partial y} (5x^2 - 3xy + xyz) + \hat{k} \frac{\partial}{\partial z} (5x^2 - 3xy + xyz) \\
 &= \hat{i}(10x - 3y + yz) + \hat{j}(-3x + xz) + \hat{k}(xy)
 \end{aligned}$$

At P (3, 4, 3),

$$\begin{aligned}
 \vec{\nabla} \phi &= \hat{i}(10x - 3y + yz) + \hat{j}(-3x + xz) + \hat{k}(xy) \\
 \vec{\nabla} \phi &= \hat{i}(10 \times 3 - 3 \times 4 + 4 \times 3) + \hat{j}(-3 \times 3 + 3 \times 3) + \hat{k}(3 \times 4) \\
 \vec{\nabla} \phi &= \hat{i}(30 - 12 + 12) + \hat{j}(-9 + 9) + \hat{k}(12) \\
 \vec{\nabla} \phi &= \hat{i}(30) + 12\hat{k}
 \end{aligned}$$

$$\text{Given, } \vec{v} = \hat{i} + \hat{j} - \hat{k}; \text{ the unit vector of } \vec{v} = \hat{a} = \frac{\vec{v}}{|\vec{v}|} = \frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{1^2 + 1^2 + (-1)^2}} = \frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$$

i)

$$\begin{aligned}
 \vec{\nabla} \phi \cdot \hat{a} &= [\hat{i}(30) + 12\hat{k}] \cdot \left[\frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}} \right] \\
 \vec{\nabla} \phi \cdot \hat{a} &= [\hat{i}(30) + 12\hat{k}] \cdot \left[\frac{\hat{i}}{\sqrt{3}} + \frac{\hat{j}}{\sqrt{3}} - \frac{\hat{k}}{\sqrt{3}} \right] \\
 \vec{\nabla} \phi \cdot \hat{a} &= \left[\frac{30}{\sqrt{3}} - \frac{12}{\sqrt{3}} \right] \\
 \vec{\nabla} \phi \cdot \hat{a} &= \frac{30 - 12}{\sqrt{3}} = \frac{18}{\sqrt{3}} = \frac{18\sqrt{3}}{3} = 6\sqrt{3}
 \end{aligned}$$

ii)

$$\vec{\nabla} \phi = \hat{i}(30) + 12\hat{k}$$

iii)

$$\left| \vec{\nabla} \phi \right| = \sqrt{(30)^2 + (12)^2} = \sqrt{900 + 144} = \sqrt{1044}$$



$$\begin{aligned}
&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a & b & c \end{vmatrix} \\
&= \hat{i} \left(\frac{\partial c}{\partial y} - \frac{\partial b}{\partial z} \right) - \hat{j} \left(\frac{\partial c}{\partial x} - \frac{\partial a}{\partial z} \right) + \hat{k} \left(\frac{\partial b}{\partial x} - \frac{\partial a}{\partial y} \right) \\
&= \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(0-0) \\
&= 0
\end{aligned}$$

Q# 46: If $\vec{v}(x,y,z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$ Find $\text{curl } \vec{V}$

Answer: The curl of a vector field $\vec{v}(x,y,z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$ is defined by

$$\begin{aligned}
\nabla \times \vec{v} &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \times (xz\hat{i} + xyz\hat{j} - y^2\hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & xyz & -y^2 \end{vmatrix} \\
&= \hat{i} \left[\frac{\partial}{\partial y}(-y^2) - \frac{\partial}{\partial z}(xyz) \right] - \hat{j} \left[\frac{\partial}{\partial x}(-y^2) - \frac{\partial}{\partial z}(xz) \right] + \hat{k} \left[\frac{\partial}{\partial x}(xyz) - \frac{\partial}{\partial y}(xz) \right] \\
&= \hat{i}[-2y - xy] - \hat{j}[0 - x] + \hat{k}[yz - 0] \\
&= -[2y + xy]\hat{i} + \hat{j}x + \hat{k}yz \\
&= -[2y + xy]\hat{i} + x\hat{j} + yz\hat{k}
\end{aligned}$$

Q# 47: What is irrotational Field or conservative vector field?

A vector field $\vec{V}$ for which the curl vanishes, that is: $\vec{V} \times \vec{V} = 0$

Q# 48: Determine $\vec{F}$ is a conservative vector field or not where

$$\vec{F} = x^2y\hat{i} + xyz\hat{j} - x^2y^2\hat{k}$$

Answer:

So all that we need to do is compute the curl and see if we get the zero vector or not.

The curl of a vector field $\vec{F} = x^2y\hat{i} + xyz\hat{j} - x^2y^2\hat{k}$ is defined by

$$\begin{aligned}\nabla \times \vec{F} &= \left(\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \times (x^2 y \hat{i} + xyz \hat{j} - x^2 y^2 \hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ x^2 y & xyz & -x^2 y^2 \end{vmatrix} \\ &= \hat{i} \left[\frac{\delta}{\delta y} (-x^2 y^2) - \frac{\delta}{\delta z} (xyz) \right] - \hat{j} \left[\frac{\delta}{\delta x} (-x^2 y^2) - \frac{\delta}{\delta z} (x^2 y) \right] \\ &\quad + \hat{k} \left[\frac{\delta}{\delta x} (xyz) - \frac{\delta}{\delta y} (x^2 y) \right] \\ &= \hat{i} [-2x^2 y - xy] - \hat{j} [-2xy^2 - 0] + \hat{k} [yz - x^2] \\ &\neq 0\end{aligned}$$

So, the curl isn't the zero vectors and so this vector field is not conservative.

Home Task

Find the values of constants a, b and c so that,

$\vec{F} = (x + 2y + az) \hat{i} + (bx - 3y - z) \hat{j} + (4x + cy + 2z) \hat{k}$ is irrotational.

Q# 49: If $\vec{A} = (xz^3 \hat{i} - 2x^2 yz \hat{j} + 2yz^4 \hat{k})$, find $\nabla \times \vec{A}$ (or curl A) at the point (1, -1, 1).

Answer:

$$\begin{aligned}\nabla \times \vec{A} &= \left(\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \times (xz^3 \hat{i} - 2x^2 yz \hat{j} + 2yz^4 \hat{k}) \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ xz^3 & -2x^2 yz & 2yz^4 \end{vmatrix} \\ &= \hat{i} \left[\frac{\delta}{\delta y} (2yz^4) - \frac{\delta}{\delta z} (-2x^2 yz) \right] - \hat{j} \left[\frac{\delta}{\delta x} (2yz^4) - \frac{\delta}{\delta z} (xz^3) \right] + \hat{k} \left[\frac{\delta}{\delta x} (-2x^2 yz) - \frac{\delta}{\delta y} (xz^3) \right] \\ &= \hat{i} [2z^4 + 2x^2 y] - \hat{j} [0 - 3xz^2] + \hat{k} [-4xyz - 0] \\ &= \hat{i} [2.1^4 + 2.1^2 \cdot (-1)] - \hat{j} [0 - 3.1.1^2] + \hat{k} [-4.1.(-1).1 - 0] \\ &= \hat{i} [2 - 2] - \hat{j} [0 - 3] + \hat{k} [4] \\ &= 3 \hat{j} + 4 \hat{k}\end{aligned}$$

Q# 55: Show that $\vec{r} \cdot \vec{r} = r^2$

Also, $\vec{r} \cdot \vec{r} = (x\hat{i} + y\hat{j} + z\hat{k}) \cdot (x\hat{i} + y\hat{j} + z\hat{k})$

$$\begin{aligned}\vec{r} \cdot \vec{r} &= x^2 + y^2 + z^2 [\because \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1; \hat{k} \cdot \hat{k} = 1] \\ &= r^2 [\because r^2 = x^2 + y^2 + z^2]\end{aligned}$$

Similarly,

$$\vec{A} \cdot \vec{A} = A^2$$

$$\vec{\nabla} \cdot \vec{\nabla} = \nabla^2$$

Q# 56: Show that, $\vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}$

Proof: L.H.S = $\vec{\nabla}$

$$\begin{aligned}&= \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \\ &= \hat{i} \frac{\partial}{\partial x} \frac{\partial r}{\partial r} + \hat{j} \frac{\partial}{\partial y} \frac{\partial r}{\partial r} + \hat{k} \frac{\partial}{\partial z} \frac{\partial r}{\partial r} \\ &= \hat{i} \frac{\partial r}{\partial x} \frac{\partial}{\partial r} + \hat{j} \frac{\partial r}{\partial y} \frac{\partial}{\partial r} + \hat{k} \frac{\partial r}{\partial z} \frac{\partial}{\partial r} \\ &= \left(\hat{i} \frac{\partial r}{\partial x} + \hat{j} \frac{\partial r}{\partial y} + \hat{k} \frac{\partial r}{\partial z} \right) \frac{\partial}{\partial r} \\ &= \left(\hat{i} \frac{x}{r} + \hat{j} \frac{y}{r} + \hat{k} \frac{z}{r} \right) \frac{\partial}{\partial r} [\because \frac{\partial r}{\partial x} = \frac{x}{r}; \frac{\partial r}{\partial y} = \frac{y}{r}; \frac{\partial r}{\partial z} = \frac{z}{r}] \\ &= \frac{x\hat{i} + y\hat{j} + z\hat{k}}{r} \frac{\partial}{\partial r} \\ &= \frac{\vec{r}}{r} \frac{\partial}{\partial r}\end{aligned}$$

Q# 57: Show that $\vec{\nabla} \cdot (\phi \vec{A}) = \phi (\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}$

Solution

Let $\vec{A} = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}$ is a vector and ϕ is a function of a variable or variables

$$\begin{aligned}\text{L.H.S } \vec{\nabla} \cdot (\phi \vec{A}) &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (\phi \vec{A}) \\ &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot \phi (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\ &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (\phi A_1 \hat{i} + \phi A_2 \hat{j} + \phi A_3 \hat{k})\end{aligned}$$

$$\begin{aligned}
\int \frac{\partial p}{p} + \int \frac{2\partial r}{r} &= \int 0 && [\text{Integrating}] \\
\ln p + 2 \ln r &= \ln A \\
\ln p + \ln r^2 &= \ln A \\
\ln pr^2 &= \ln A && [\because \ln ab = \ln a + \ln b] \\
pr^2 &= A \\
p &= Ar^{-2} \\
\frac{\partial f(r)}{\partial r} &= Ar^{-2} \\
\int \frac{\partial f(r)}{\partial r} \partial r &= \int Ar^{-2} \partial r \\
\int \frac{\partial}{\partial r} \{f(r)\} \partial r &= \int Ar^{-2} \partial r
\end{aligned}$$

Integrating

$$f(r) = A \frac{r^{-2+1}}{-2+1} + B$$

$$f(r) = A \frac{r^{-1}}{-1} + B$$

$$f(r) = -A \frac{1}{r} + B$$

$$f(r) = \frac{-A}{r} + B$$

$$f(r) = B + \frac{c}{r}, \quad \text{Where, } c = -A$$

Q# 66: Show that $\nabla^2 \left(\frac{1}{r} \right) = 0$

$$\text{L.H.S} = \nabla^2 \left(\frac{1}{r} \right)$$

$$= \vec{\nabla} \cdot \vec{\nabla} \left(\frac{1}{r} \right) \quad [\because \vec{\nabla} \cdot \vec{\nabla} = \nabla^2]$$

$$= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} \left(\frac{1}{r} \right) \right] \quad [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}]$$

$$= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{-1}) \right]$$

$$= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} (-1) r^{-1-1} \right]$$

$$\begin{aligned}
&= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} (-1) r^{-2} \right] \\
&= \vec{\nabla} \cdot \left[\vec{r} (-1) r^{-2} r^{-1} \right] \\
&= \vec{\nabla} \cdot \left[\vec{r} (-1) r^{-3} \right] \\
&= \vec{\nabla} \cdot \left[-r^{-3} \vec{r} \right] \\
&= \vec{\nabla} \cdot \left[-\frac{\vec{r}}{r^3} \right] \\
&= \vec{\nabla} \cdot \left[-\vec{r} r^{-3} \right] \\
&= \vec{\nabla} \cdot \left[-r^{-3} \vec{r} \right] \\
&= -r^{-3} (\vec{\nabla} \cdot \vec{r}) - (\vec{\nabla} r^{-3}) \cdot \vec{r} \\
&= -3r^{-3} - \left\{ \frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{-3}) \right\} \cdot \vec{r} \\
&= -\frac{3}{r^3} - \left\{ \frac{1}{r} (-3) r^{-3-1} \right\} (\vec{r} \cdot \vec{r}) \\
&= -\frac{3}{r^3} - \left\{ \frac{1}{r} (-3) r^{-4} \right\} (\vec{r} \cdot \vec{r}) \\
&= -\frac{3}{r^3} - \left\{ (-3) r^{-4} r^{-1} \right\} (\vec{r} \cdot \vec{r}) \\
&= -\frac{3}{r^3} - \left\{ (-3) r^{-4-1} \right\} (\vec{r} \cdot \vec{r}) \\
&= -\frac{3}{r^3} - \left\{ (-3) r^{-5} \right\} (\vec{r} \cdot \vec{r}) \\
&= -\frac{3}{r^3} - \left\{ \frac{-3}{r^5} \right\} (\vec{r} \cdot \vec{r}) \\
&= -\frac{3}{r^3} - \left\{ \frac{-3}{r^5} \right\} r^2 \\
&= -\frac{3}{r^3} + \frac{3}{r^3} = 0
\end{aligned}$$

$$[\because \vec{\nabla} \cdot (\phi \vec{A}) = \phi (\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}]$$

$$[\because \vec{\nabla} \cdot \vec{r} = 3] \& [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}]$$

$$[\because \vec{r} \cdot \vec{r} = r^2]$$

Q# 67: Find $\vec{\nabla} \phi$ if (a) $\phi = \ln \left| \vec{r} \right|$ (b) $\phi = \frac{1}{\left| \vec{r} \right|}$

Answer:

a) Given $\phi = \ln \left| \vec{r} \right|$

So we can rewrite the integral above: $\int_{AB} \vec{F} d\vec{s} = \int_{AB} P dx + Q dy$

Because,

The work done by $\vec{F}$ in moving the particle from the tail to the head $d\vec{S}$ is approximately:

$$\vec{F} \cdot d\vec{S} \text{ -----(i)}$$

If $\vec{F}(x,y) = P(x,y)\hat{i} + Q(x,y)\hat{j}$ (Where P and Q are functions of x and y)

$$\text{and } d\vec{S} = dx\hat{i} + dy\hat{j}$$

$$\text{Then, } \vec{F} \cdot d\vec{S} = (P\hat{i} + Q\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$$

$$\text{Then, } \vec{F} \cdot d\vec{S} = P dx + Q dy \quad [\because \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1] \text{ -----(ii)}$$

Since $P dx + Q dy$ is a local estimate (from the tail to the head $d\vec{S}$) of the work, the total work is represented by a line integral:

$$\text{Work} = \int_{AB} \vec{F} \cdot d\vec{S} = \int_{AB} (P\hat{i} + Q\hat{j}) \cdot (dx\hat{i} + dy\hat{j}) = \int_{AB} (P dx + Q dy) \text{ -----(iii)}$$

In three dimension,

$$\text{Work} = \int_{AB} \vec{F} \cdot d\vec{S} = \int_{AB} (P\hat{i} + Q\hat{j} + R\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = \int_{AB} (P dx + Q dy + R dz) \text{ ----}$$

(iv)

Q # 75: How much work is accomplished by the force $\vec{F}(x,y) = xy\hat{i} + y\hat{j}$ in pushing a particle from (0,0) to (3,9) along the parabola $y = x^2$?

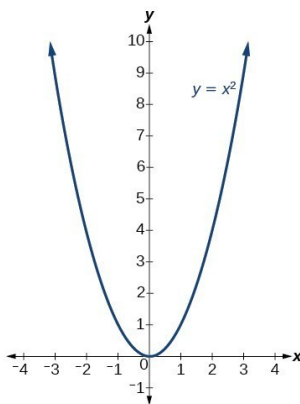


Figure # 86

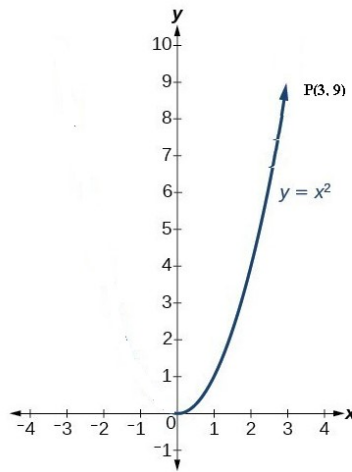


Figure # 87

Answer:

Figure shows the path of the particle, Call this path OP. Then

$$\text{Here, } \vec{F}(x, y) = xy \hat{i} + y \hat{j}, \text{-----(i)}$$

We know,

$$\vec{F}(x, y) = P \hat{i} + Q \hat{j} \text{-----(ii)}$$

Comparing (i) and (ii),

$$P = xy, Q = y$$

We know,

$$\text{Work} = \int_{OP} \vec{F} \cdot d\vec{S} = \int_{OP} (Pdx + Qdy)$$

$$\text{Work} = \int_{OP} (xydx + ydy) \quad [\because P = xy, Q = y]$$

To evaluate this integral, let us use x as the parameter,

Then, Given $y = x^2$

$$\frac{dy}{dx} = \frac{d}{dx}(x^2)$$

$$\frac{dy}{dx} = 2x$$

$$\therefore dy = 2xdx$$

$$\text{Work} = \int_0^3 (xydx + ydy)$$

$$\text{Work} = \int_0^3 [(x \cdot x^2 dx + x^2 \cdot 2x dx)]$$

$$\text{Work} = \int_0^3 (x^3 dx + 2x^3 dx)$$

$$\text{Work} = \int_0^3 3x^3 dx = 3 \left[\frac{x^{3+1}}{3+1} \right]_0^3 = \frac{3}{4} [x^4]_0^3 = \frac{3}{4} \times 3^4 = \frac{243}{4} \text{ Answer.}$$

Q # 76: Evaluate $\int_C xy dx$ from **B(1,0)** to **C(0,1)** along the curve C that is the portion of $x^2 + y^2 = 1$ in the first quadrant.

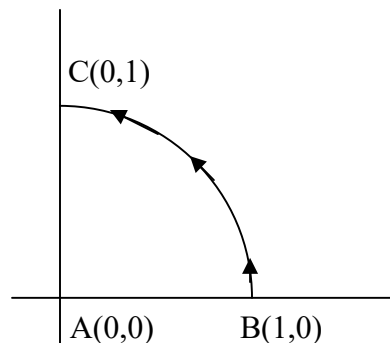


Figure # 88

$$\therefore \frac{dz}{dt} = 3t^2$$

$$\therefore dz = 3t^2 dt$$

From (i),

$$\vec{F} \cdot d\vec{r} = 3xydx - 5zdy + 10xdz$$

$$\vec{F} \cdot d\vec{r} = 3(t^2 + 1)(2t^2)2tdt - 5t^3 4tdt + 10(t^2 + 1)3t^2 dt$$

$$\vec{F} \cdot d\vec{r} = 3(t^2 + 1)(4t^3)dt - 20t^4 dt + 10(t^2 + 1)3t^2 dt$$

$$\vec{F} \cdot d\vec{r} = 3(4t^5 + 4t^3)dt - 20t^4 dt + 10(3t^4 + 3t^2)dt$$

$$\vec{F} \cdot d\vec{r} = (12t^5 + 12t^3)dt - 20t^4 dt + (30t^4 + 30t^2)dt$$

$$\vec{F} \cdot d\vec{r} = 12t^5 dt + 12t^3 dt - 20t^4 dt + 30t^4 dt + 30t^2 dt$$

$$\vec{F} \cdot d\vec{r} = 12t^5 dt + 10t^4 dt + 12t^3 dt + 30t^2 dt$$

$$\text{Total work done} = \int_1^2 \vec{F} \cdot d\vec{r} = \int_1^2 (12t^5 dt + 10t^4 dt + 12t^3 dt + 30t^2 dt)$$

$$= \int_1^2 12t^5 dt + \int_1^2 10t^4 dt + \int_1^2 12t^3 dt + \int_1^2 30t^2 dt$$

$$= \left[12 \frac{t^6}{6} + 10 \frac{t^5}{5} + 12 \frac{t^4}{4} + 30 \frac{t^3}{3} \right]_1^2$$

$$= [2t^6 + 2t^5 + 3t^4 + 10t^3]_1^2$$

$$= (2 \times 2^6 + 2 \times 2^5 + 3 \times 2^4 + 10 \times 2^3) - (2 \times 1^6 + 2 \times 1^5 + 3 \times 1^4 + 10 \times 1^3)$$

$$= (2 \times 64 + 2 \times 32 + 3 \times 16 + 10 \times 8) - (2 + 2 + 3 + 10)$$

$$= (128 + 64 + 48 + 80) - (2 + 2 + 3 + 10)$$

$$= (320) - (17)$$

$$= 303$$

 Q # 85: If $\vec{F} = xy\hat{i} - z\hat{j} + x^2\hat{k}$ and C is the curve, $x = t^2$, $y = 2t$, $z = t^3$ from

$t = 0$ to $t = 1$, then evaluate the line integral, $\int_C \vec{F} \cdot d\vec{r}$

Answer: We have the position vector

$$\therefore \vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \quad [\text{Page no 48, Figure no 57, Equation no (i)}]$$

$$\therefore \vec{dr} = dx \hat{i} + dy \hat{j} + dz \hat{k}$$

Given

$$x = t^2$$

$$\therefore \frac{dx}{dt} = \frac{d}{dt}(t^2)$$

$$\therefore \frac{dx}{dt} = 2t$$

$$\therefore dx = 2t dt$$

$$y = 2t$$

$$\therefore \frac{dy}{dt} = \frac{d}{dt}(2t)$$

$$\therefore \frac{dy}{dt} = 2$$

$$\therefore dy = 2 dt$$

$$z = t^3$$

$$\therefore \frac{dz}{dt} = \frac{d}{dt}(t^3)$$

$$\therefore \frac{dz}{dt} = 3t^2$$

$$\therefore dz = 3t^2 dt$$

$$\therefore \vec{F} = xy\hat{i} - z\hat{j} + x^2\hat{k}$$

$$\Rightarrow \vec{F} = (t^2 \times 2t)\hat{i} - t^3\hat{j} + (t^2)^2\hat{k}$$

$$\Rightarrow \vec{F} = 2t^3\hat{i} - t^3\hat{j} + t^4\hat{k} \text{ -----(i)}$$

$$\therefore \vec{dr} = dx \hat{i} + dy \hat{j} + dz \hat{k}$$

$$\therefore \vec{dr} = 2t dt \hat{i} + 2dt \hat{j} + 3t^2 dt \hat{k} \text{ -----(ii)}$$

From (i) and (ii)

$$\vec{F} \times \vec{dr} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2t^3 & -t^3 & t^4 \\ 2t dt & 2dt & 3t^2 dt \end{vmatrix}$$

$$\begin{aligned}\vec{F} \times \vec{dr} &= (-3t^5 - 2t^4)dt\hat{i} - (6t^5 - 2t^5)dt\hat{j} + (4t^3 + 2t^4)dt\hat{k} \\ \vec{F} \times \vec{dr} &= (-3t^5 - 2t^4)dt\hat{i} - (4t^5)dt\hat{j} + (4t^3 + 2t^4)dt\hat{k} \\ \int_c \vec{F} \times \vec{dr} &= \int_0^1 (-3t^5 - 2t^4)dt\hat{i} - (4t^5)dt\hat{j} + (4t^3 + 2t^4)dt\hat{k} \\ \int_c \vec{F} \times \vec{dr} &= \left[-3\frac{t^6}{6} - 2\frac{t^5}{5}\right]_0^1\hat{i} - \left[4\frac{t^6}{6}\right]_0^1\hat{j} + \left[4\frac{t^4}{4} + 2\frac{t^5}{5}\right]_0^1\hat{k} \\ \int_c \vec{F} \times \vec{dr} &= \left[-3\frac{1^6}{6} - 2\frac{1^5}{5}\right]\hat{i} - \left[4\frac{1^6}{6}\right]\hat{j} + \left[4\frac{1^4}{4} + 2\frac{1^5}{5}\right]\hat{k} \\ \int_c \vec{F} \times \vec{dr} &= \left[-\frac{1}{2} - \frac{2}{5}\right]\hat{i} - \left[\frac{2}{3}\right]\hat{j} + \left[1 + \frac{2}{5}\right]\hat{k} \\ \int_c \vec{F} \times \vec{dr} &= -\frac{9}{10}\hat{i} - \frac{2}{3}\hat{j} + \frac{7}{5}\hat{k} \text{ Answer}\end{aligned}$$

Q # 86 :Home Task:

11. Find the work done in moving a particle once around a circle C in the xy plane, If the circle has center at the origin and radius 3 and the force field is given by

$$\vec{F} = (2x - y + z)\hat{i} + (x + y - z^2)\hat{j} + (3x - 2y + 4z)\hat{k}$$

Hints:

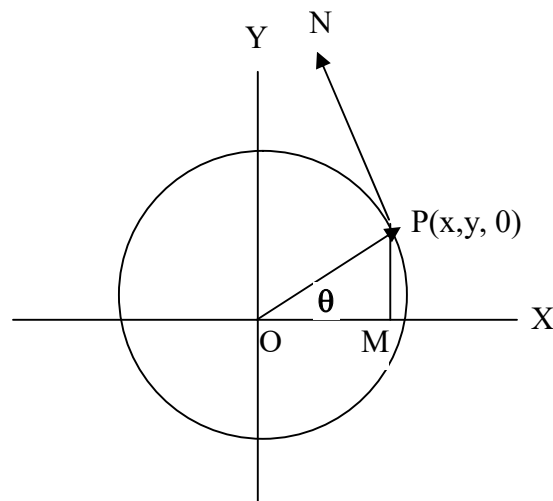


Figure # 101

Answer: Let the equation of the circle is : $x^2 + y^2 = 3^2$

Let,

$$\angle POM = \theta, \quad OP = 3$$

$$\frac{PM}{OP} = \sin \theta$$

$$\frac{y}{3} = \sin \theta$$

$$\Rightarrow y = 3 \sin \theta$$

$$\Rightarrow dy = 3 \cos \theta d\theta$$

Similarly,

$$\frac{OM}{OP} = \cos \theta$$

$$\frac{x}{3} = \cos \theta$$

$$\Rightarrow x = 3 \cos \theta$$

$$\Rightarrow dx = -3 \sin \theta d\theta$$

Since the circle is in two dimensional, hence, $z = 0 \Rightarrow dz = 0$

Let, the position Vector is

$$\vec{r}(x, y, z) = x \hat{i} + y \hat{j} + z \hat{k}$$

[Page no 48, Figure no 57, Equation no (i)]

$$\therefore d\vec{r} = dx \hat{i} + dy \hat{j} + dz \hat{k}$$

$$\text{Let, } \vec{F} = F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k}$$

$$\vec{F} \cdot d\vec{r} = (F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k}) \cdot (dx \hat{i} + dy \hat{j} + dz \hat{k})$$

$$\vec{F} \cdot d\vec{r} = F_1 dx + F_2 dy + F_3 dz$$

$$\vec{F} \cdot d\vec{r} = F_1 dx + F_2 dy + F_3 dz$$

$$\text{Given, } \vec{F} = (2x - y + z) \hat{i} + (x + y - z^2) \hat{j} + (3x - 2y + 4z) \hat{k}$$

$$\text{So, } \vec{F} \cdot d\vec{r} = F_1 dx + F_2 dy + F_3 dz$$

$$\vec{F} \cdot d\vec{r} = (2x - y + z)dx + (x + y - z^2)dy + (3x - 2y + 4z)dz$$

$$\vec{F} \cdot d\vec{r} = (2x - y)dx + (x + y)dy + (3x - 2y) \cdot 0 \quad [z=0, dz=0]$$

$$\vec{F} \cdot d\vec{r} = (2x - y)dx + (x + y)dy \quad [z=0]$$

$$\begin{aligned}
&= \int_0^{2\pi} \vec{F} \cdot d\vec{r} \\
\text{Hence, total work} &= \int_0^{2\pi} (2x - y)dx + (x + y)dy \\
&= \int_0^{2\pi} \{(2.3 \cos \theta - 3 \sin \theta)(-3 \sin \theta)d\theta + (3 \cos \theta + 3 \sin \theta)3 \cos \theta d\theta\} \\
&= \dots\dots\dots
\end{aligned}$$

02. Find the work done in moving a particle once around a circle C in the xy plane, if the circle has a centre (0, 0) and radius 1 and if the force field $\vec{F}$ is given by $\vec{F} = (2x - y + 2z)\hat{i} + (x + y - z)\hat{j} + (3x - 2y - 5z)\hat{k}$

Answer: Let the equation of the circle is $x^2 + y^2 = 1$

Let, $x = r \cos \theta$ & $y = r \sin \theta$

Here, radius $r = 1$

$x = \cos \theta$ $y = \sin \theta$

then, $dx = -\sin \theta d\theta$ and $dy = \cos \theta d\theta$

since the circle is in two dimensional, hence, $z = 0 \Rightarrow dz = 0$

let, the position Vector is

$$\vec{r}(x, y, z) = x\hat{i} + y\hat{j} + z\hat{k}$$

[Page no 48, Figure no 57, Equation no (i)]

$$\therefore d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$\text{Let, } \vec{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$$

$$\vec{F} \cdot d\vec{r} = (F_1\hat{i} + F_2\hat{j} + F_3\hat{k}) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

$$\vec{F} \cdot d\vec{r} = F_1 dx + F_2 dy + F_3 dz$$

$$\vec{F} \cdot d\vec{r} = F_1 dx + F_2 dy + F_3 dz$$

$$\text{Given, } \vec{F} = (2x - y + 2z)\hat{i} + (x + y - z)\hat{j} + (3x - 2y - 5z)\hat{k}$$

$$\text{So, } \vec{F} \cdot d\vec{r} = F_1 dx + F_2 dy + F_3 dz$$

$$\vec{F} \cdot d\vec{r} = (2x - y + 2z)dx + (x + y - z)dy + (3x - 2y - 5z)dz$$

$$\vec{F} \cdot d\vec{r} = (2x - y + 2z)dx + (x + y - z)dy + 0 \quad [dz = 0]$$

$$\vec{F} \cdot d\vec{r} = (2x - y)dx + (x + y)dy \quad [z = 0]$$

$$\begin{aligned}
&= \int_0^{2\pi} \vec{F} \cdot d\vec{r} \\
\text{Hence, total work} &= \int_0^{2\pi} \vec{F} \cdot d\vec{r}
\end{aligned}$$

$$\begin{aligned}
&= \left[9z\hat{i} + 6\frac{z^2}{2}\hat{j} - 8z\hat{k} \right]_0^4 \\
&= \left[9z\hat{i} + 3z^2\hat{j} - 8z\hat{k} \right]_0^4 \\
&= \left[(9 \times 4\hat{i} + 3 \times 4^2\hat{j} - 8 \times 4\hat{k}) - (9 \times 0\hat{i} + 3 \times 0^2\hat{j} - 8 \times 0\hat{k}) \right] \\
&= \left[36\hat{i} + 3 \times 16\hat{j} - 32\hat{k} \right] \\
&= 4 \left[9\hat{i} + 12\hat{j} - 8\hat{k} \right]
\end{aligned}
\quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1}; \int dz = z \right]$$

✓ Q # 90: Show that $\iint_S \vec{F} \cdot \hat{\eta} ds = \frac{3}{2}$; Where $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ and S is the surface of the cube bounded by the planes $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$

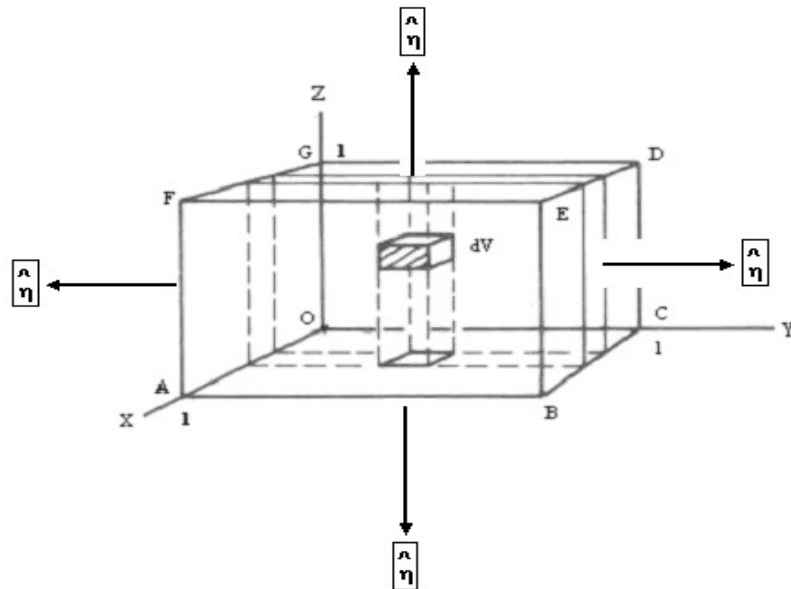


Figure # 113

Serial no	Surface	$\hat{\eta}=?$	ds	Plane
1	OABC	$\hat{\eta} = -\hat{k}$	$dx dy$	$z = 0$
2	DEFG	$\hat{\eta} = \hat{k}$	$dx dy$	$z = 1$
3	OAFG	$\hat{\eta} = -\hat{j}$	$dx dz$	$y = 0$
4	BCDE	$\hat{\eta} = \hat{j}$	$dx dz$	$y = 1$
5	OCDG	$\hat{\eta} = -\hat{i}$	$dy dz$	$x = 0$
6	ABEF	$\hat{\eta} = \hat{i}$	$dy dz$	$x = 1$

Now,

$$\iint_S \vec{F} \cdot \hat{n} ds = \iint_{OABC} \vec{F} \cdot \hat{n} ds + \iint_{DEFG} \vec{F} \cdot \hat{n} ds + \iint_{OAFG} \vec{F} \cdot \hat{n} ds + \iint_{BCDE} \vec{F} \cdot \hat{n} ds + \iint_{OCDG} \vec{F} \cdot \hat{n} ds + \iint_{ABEF} \vec{F} \cdot \hat{n} ds \quad (1)$$

$$1. \iint_{OABC} \vec{F} \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (-\hat{k}) dx dy$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \int_0^1 -yz dx dy$$

$$= - \int_0^1 \int_0^1 yz dx dy = - \int_0^1 [yzx]_0^1 dy \quad [\because \int dx = x]$$

$$= - \int_0^1 [yz \times 1 - yz \times 0] dy = - \int_0^1 [yz(1 - 0)] dy$$

$$= - \left[z \frac{y^2}{2} \right]_0^1 = - \left[z \frac{1^2}{2} - z \frac{0^2}{2} \right] = - \frac{z}{2} (1 - 0) = - \frac{z}{2} = 0 [\because z = 0]$$

2.

$$\iint_{DEFG} \vec{F} \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (\hat{k}) dx dy = \int_0^1 \int_0^1 yz dx dy$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \int_0^1 yz dx dy = \int_0^1 [yzx]_0^1 dy = \int_0^1 [yz \times 1 - yz \times 0] dy \quad [\because \int dx = x]$$

$$= \int_0^1 [yz] dy = \left[z \frac{y^2}{2} \right]_0^1 = \left[z \frac{1^2}{2} - z \frac{0^2}{2} \right] = \frac{z}{2} (1 - 0) = \frac{z}{2} = \frac{1}{2} [\because z = 1]$$

3.

$$\iint_{OAFG} \vec{F} \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (-\hat{j}) dx dz = \int_0^1 \int_0^1 y^2 dx dz$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \int_0^1 y^2 dx dz = \int_0^1 [xy^2]_0^1 dz \quad [\because \int dx = x]$$

$$= \int_0^1 [1 \times y^2 - 0 \times y^2] dz$$

$$= \int_0^1 [y^2 - 0] dz = [y^2 z]_0^1 = [y^2 \times 1 - y^2 \times 0] \quad [\because \int dz = z]$$

$$= y^2 (1 - 0) = y^2 = 0 \quad [\because y = 0]$$

4.

$$\begin{aligned}
 \iint_{BCDE} \vec{F} \cdot \hat{n} ds &= \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (\hat{j}) dx dz = \int_0^1 \int_0^1 -y^2 dx dz \\
 [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0] \\
 &= -\int_0^1 \int_0^1 y^2 dx dz = -\int_0^1 [xy^2]_0^1 dz = -\int_0^1 [1 \times y^2 - 0 \times y^2] dz \quad [\because \int dx = x] \\
 &= -\int_0^1 [y^2 - 0] dz = -[y^2 z]_0^1 \quad [\because \int dz = z] \\
 &= -[y^2 \times 1 - y^2 \times 0] \\
 &= -y^2 (1 - 0) = -y^2 = -1 \quad [\because y = 1]
 \end{aligned}$$

$$\begin{aligned}
 5. \iint_{OCDE} \vec{F} \cdot \hat{n} ds &= \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (-\hat{i}) dy dz \\
 [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0] \\
 &= \int_0^1 \int_0^1 -4xz dy dz \\
 &= -\int_0^1 \int_0^1 4xz dy dz = -\int_0^1 [4xyz]_0^1 dz = -\int_0^1 [4x \times 1 \times z - 4x \times 0 \times z] dz \quad [\because \int dy = y] \\
 &= -\int_0^1 [4xz(1 - 0)] dz = -\left[4x \frac{z^2}{2}\right]_0^1 = -\left[4x \frac{1^2}{2} - 4x \frac{0^2}{2}\right] \\
 &= -4x \left(\frac{1}{2} - 0\right) = -2x = 0 \quad [\because x = 0]
 \end{aligned}$$

$$\begin{aligned}
 6. \iint_{ABEF} \vec{F} \cdot \hat{n} ds &= \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot \hat{n} ds = \iint_S (4xz \hat{i} - y^2 \hat{j} + yz \hat{k}) \cdot (\hat{i}) dy dz \\
 [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0] \\
 &= \int_0^1 \int_0^1 4xz dy dz \\
 &= \int_0^1 \int_0^1 4xz dy dz = \int_0^1 [4xyz]_0^1 dz = \int_0^1 [4x \times 1 \times z - 4x \times 0 \times z] dz \quad [\because \int dy = y] \\
 &= \int_0^1 [4xz(1 - 0)] dz = \left[4x \frac{z^2}{2}\right]_0^1 = \left[4x \frac{1^2}{2} - 4x \frac{0^2}{2}\right] \\
 &= 4x \left(\frac{1}{2} - 0\right) = 2x = 2 \quad [\because x = 1]
 \end{aligned}$$

Putting the values in (1),

Now,

$$\iint_S \vec{F} \cdot \hat{n} ds = \iint_{OABC} \vec{F} \cdot \hat{n} ds + \iint_{DEFG} \vec{F} \cdot \hat{n} ds + \iint_{OAFG} \vec{F} \cdot \hat{n} ds + \iint_{BCDE} \vec{F} \cdot \hat{n} ds + \iint_{OCDG} \vec{F} \cdot \hat{n} ds + \iint_{ABEF} \vec{F} \cdot \hat{n} ds$$

$$\iint_S \vec{F} \cdot \hat{n} ds = 0 + \frac{1}{2} + 0 - 1 + 0 + 2 = \frac{1}{2} + 1 = \frac{3}{2} \text{ (Proved)}$$

Area Enclosed by a closed Curve:

One of the earliest applications of integration is finding the area of a plane figure bounded by the x-axis, the curve $y = f(x)$ and ordinates at $x = x_1$ and $x = x_2$.

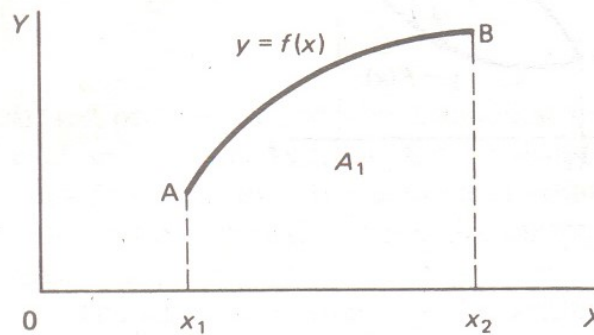


Figure # 114

$$A_1 = \int_{x_1}^{x_2} y dx = \int_{x_1}^{x_2} f(x) dx \text{ -----(i)}$$

If points A and B are joined by another curve $y = F(x)$

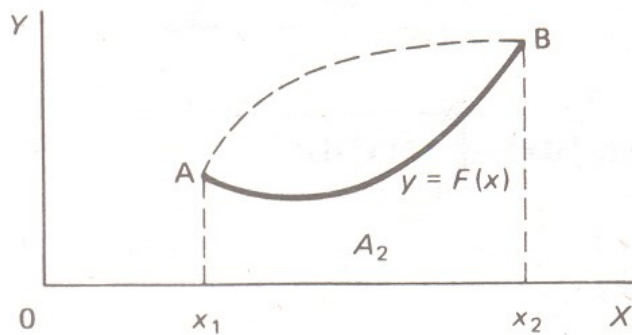


Figure # 115

$$A_2 = \int_{x_1}^{x_2} y dx = \int_{x_1}^{x_2} F(x) dx \text{ -----(ii)}$$

Combining the two figures, we have

OA : c_1 is the line $y = 0 \therefore dy = 0$, Then

$I_1 = \oint (x^2 dx - 2xy dy)$ for this part becomes

$$I_1 = \int_0^1 (x^2 dx - 2x \cdot 0 \cdot 0) = \int_0^1 x^2 dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3}$$

b) AB : c_2 is the line: $\frac{y-0}{0-1} = \frac{x-1}{1-0} \Rightarrow \frac{y}{-1} = \frac{x-1}{1} \Rightarrow y = -x + 1$

$$\therefore \frac{dy}{dx} = -1 + 0 \Rightarrow dy = -dx$$

Then

$I_2 = \oint (x^2 dx - 2xy dy)$ for this part becomes

$$I_2 = \int_1^0 \{x^2 dx - 2x(-x+1)(-dx)\} = \int_1^0 (x^2 - 2x^2 + 2x)(dx) = \int_1^0 (2x - x^2) dx = \left[\frac{2x^2}{2} - \frac{x^3}{3} \right]_1^0 = -\frac{2}{3}$$

c) BO : c_3 is the line $x = 0 \therefore dx = 0$, Then

$I_3 = \oint (x^2 dx - 2xy dy)$ for this part becomes

$$I_3 = \oint (0^2 \cdot 0 - 2 \cdot 0 \cdot y dy) = 0$$

$$\text{Finally } I = I_1 + I_2 + I_3 = \frac{1}{3} - \frac{2}{3} + 0 = -\frac{1}{3}$$

Q # 95: Evaluate the area of a circle $x^2 + y^2 = 4$.

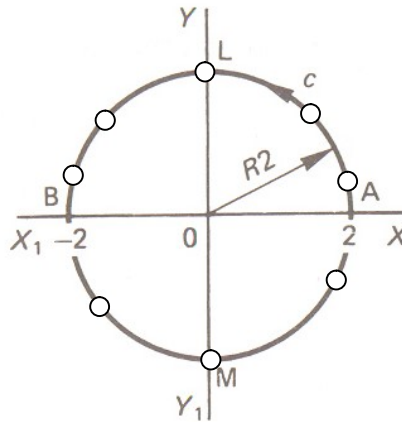


Figure # 127

Answer: Given $x^2 + y^2 = 4$

$$\Rightarrow y^2 = 4 - x^2$$

$$\Rightarrow y = \pm \sqrt{4 - x^2}$$

The Equation of the upper curve ALB: $y = +\sqrt{4 - x^2}$ between $x = 2$ and $x = -2$

And The equation of the lower curve BMA : $y = -\sqrt{4-x^2}$ between $x = -2$ and $x = 2$

We have, Area: $A = -\oint y dx = -\left\{ \int_{x_1}^{x_2} F(x) dx + \int_{x_2}^{x_1} f(x) dx \right\}$

$$\therefore A = -\left[\int_{-2}^2 \{-(\sqrt{4-x^2})\} dx + \int_2^{-2} \sqrt{4-x^2} dx \right]$$

$$\therefore A = -\left[\int_{-2}^2 \sqrt{4-x^2} dx + \int_{-2}^2 \{-(\sqrt{4-x^2})\} dx \right]$$

$$\therefore A = -\left[\int_{-2}^2 \sqrt{4-x^2} dx - \int_{-2}^2 \sqrt{4-x^2} dx \right]$$

$$\therefore A = -\left[\int_{-2}^2 \sqrt{4-x^2} dx + \int_{-2}^2 \sqrt{4-x^2} dx \right]$$

$$\therefore A = -\left[2 \int_{-2}^2 \sqrt{4-x^2} dx \right]$$

$$\therefore A = (-2)(-1) \int_{-2}^2 \sqrt{4-x^2} dx$$

$$\therefore A = 2 \int_{-2}^2 \sqrt{4-x^2} dx \text{ -----(i)}$$

$$\therefore A = 2 \times 2 \int_0^2 \sqrt{4-x^2} dx \quad \left[\because \int_{-a}^{+a} f(x) dx = 2 \int_0^a f(x) dx \right]$$

$$\therefore A = 4 \int_0^2 \sqrt{4-x^2} dx \text{ -----(ii)}$$

Let, $x = 2 \sin \theta$

$$\Rightarrow \frac{dx}{d\theta} = \frac{d}{d\theta} (2 \sin \theta)$$

$$\Rightarrow \frac{dx}{d\theta} = 2 \cos \theta$$

$$\Rightarrow dx = 2 \cos \theta d\theta$$

$$\text{Now, } \sqrt{4-x^2} = \sqrt{4-(2 \sin \theta)^2} = \sqrt{4-4 \sin^2 \theta} = \sqrt{4(1-\sin^2 \theta)} = \sqrt{4 \cos^2 \theta} = 2 \cos \theta$$

Given, $x = 2 \sin \theta$

$$\Rightarrow \theta = \sin^{-1}(x/2)$$

x	0	2
$\theta = \sin^{-1}(x/2)$	$\theta = \sin^{-1}(0/2)$ $\Rightarrow \theta = \sin^{-1}(0)$ $\Rightarrow \theta = \sin^{-1}(0)$	$\theta = \sin^{-1}(2/2)$ $\Rightarrow \theta = \sin^{-1}(1)$ $\Rightarrow \theta = \sin^{-1}(1)$

	$\Rightarrow \theta = \sin^{-1} \sin \theta$ $\Rightarrow \theta = 0$	$\Rightarrow \theta = \sin^{-1}(\sin \frac{\pi}{2})$ $\Rightarrow \theta = \frac{\pi}{2}$
--	---	---

From (ii),

$$\therefore A = 4 \int_0^2 \sqrt{4 - x^2} dx$$

$$\therefore A = 4 \int_0^{\pi/2} \sqrt{4 - (2 \sin \theta)^2} 2 \cos \theta d\theta$$

$$\therefore A = 4 \int_0^{\pi/2} \sqrt{4 - 4 \sin^2 \theta} 2 \cos \theta d\theta$$

$$\therefore A = 4 \int_0^{\pi/2} \sqrt{4(1 - \sin^2 \theta)} 2 \cos \theta d\theta$$

$$\therefore A = 4 \int_0^{\pi/2} \sqrt{4 \cos^2 \theta} 2 \cos \theta d\theta$$

$$\therefore A = 4 \int_0^{\pi/2} 2 \cos \theta \times 2 \cos \theta d\theta$$

$$\therefore A = 4 \int_0^{\pi/2} 4 \cos^2 \theta d\theta$$

$$\therefore A = 4 \times 2 \int_0^{\pi/2} 2 \cos^2 \theta d\theta$$

$$\therefore A = 8 \int_0^{\pi/2} 2 \cos^2 \theta d\theta$$

$$\therefore A = 8 \int_0^{\pi/2} (1 + \cos 2\theta) d\theta$$

$$[[2 \cos^2 \theta = 1 + \cos 2\theta]]$$

$$\therefore A = 8 \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\pi/2}$$

$$\therefore A = 8 \left[\left(\frac{\pi}{2} - 0 \right) + \left(\frac{\sin 2 \times \frac{\pi}{2}}{2} - 0 \right) \right]$$

$$\therefore A = 8 \left[\frac{\pi}{2} + \frac{\sin \pi}{2} \right]$$

$$\therefore A = 8 \left[\frac{\pi}{2} + \frac{0}{2} \right]$$

$$\therefore A = 8 \left[\frac{\pi}{2} \right]$$

$$\therefore A = 4\pi \quad \text{Answer}$$

Or from (i),

$$\therefore A = 2 \int_{-2}^2 \sqrt{4 - x^2} dx \text{ -----(i)}$$

$$\text{Let, } x = 2 \sin \theta$$

$$\Rightarrow \frac{dx}{d\theta} = \frac{d}{d\theta} (2 \sin \theta)$$

$$\Rightarrow \frac{dx}{d\theta} = 2 \cos \theta$$

$$\Rightarrow dx = 2 \cos \theta d\theta$$

$$\text{Now, } \sqrt{4 - x^2} = \sqrt{4 - (2 \sin \theta)^2} = \sqrt{4 - 4 \sin^2 \theta} = \sqrt{4(1 - \sin^2 \theta)} = \sqrt{4 \cos^2 \theta} = 2 \cos \theta$$

$$\text{Given, } x = 2 \sin \theta$$

$$\frac{x}{2} = \sin \theta$$

$$\sin \theta = \frac{x}{2}$$

$$\Rightarrow \theta = \sin^{-1} \left(\frac{x}{2} \right)$$

x	- 2	2
$\theta = \sin^{-1} \left(\frac{x}{2} \right)$	$\theta = \sin^{-1} \left(\frac{-2}{2} \right)$ $\Rightarrow \theta = \sin^{-1} (-1)$ $\Rightarrow \theta = \sin^{-1} (-1)$ $\Rightarrow \theta = -\sin^{-1} (1)$ $\Rightarrow \theta = -\sin^{-1} \left(\sin \frac{\pi}{2} \right)$ $\Rightarrow \theta = -\frac{\pi}{2}$	$\theta = \sin^{-1} \left(\frac{2}{2} \right)$ $\Rightarrow \theta = \sin^{-1} (1)$ $\Rightarrow \theta = \sin^{-1} (1)$ $\Rightarrow \theta = \sin^{-1} \left(\sin \frac{\pi}{2} \right)$ $\Rightarrow \theta = \frac{\pi}{2}$

From (i),

$$\therefore A = 2 \int_{-2}^2 \sqrt{4 - x^2} dx$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} \sqrt{4 - (2 \sin \theta)^2} 2 \cos \theta d\theta$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} \sqrt{4 - 4 \sin^2 \theta} 2 \cos \theta d\theta$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} \sqrt{4(1 - \sin^2 \theta)} 2 \cos \theta d\theta$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} \sqrt{4 \cos^2 \theta} 2 \cos \theta d\theta$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} 2 \cos \theta \times 2 \cos \theta d\theta$$

$$\therefore A = 2 \int_{-\pi/2}^{\pi/2} 4 \cos^2 \theta d\theta$$

$$\therefore A = 2 \times 2 \int_{-\pi/2}^{\pi/2} 2 \cos^2 \theta d\theta$$

$$\therefore A = 4 \int_{-\pi/2}^{\pi/2} 2 \cos^2 \theta d\theta$$

$$\therefore A = 4 \int_{-\pi/2}^{\pi/2} (1 + \cos 2\theta) d\theta \quad [2 \cos^2 \theta = 1 + \cos 2\theta]$$

$$\therefore A = 4 \int_{-\pi/2}^{\pi/2} 1 d\theta + 4 \int_{-\pi/2}^{\pi/2} \cos 2\theta d\theta$$

$$\therefore A = 4[\theta]_{-\pi/2}^{\pi/2} + 4 \left[\frac{\sin 2\theta}{2} \right]_{-\pi/2}^{\pi/2}$$

$$\therefore A = 4 \left[\left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) \right] + 4 \left[\left(\frac{\sin 2 \times \frac{\pi}{2}}{2} - \frac{\sin 2 \times \left(-\frac{\pi}{2} \right)}{2} \right) \right]$$

$$\therefore A = 4 \left[\left(\frac{\pi}{2} + \frac{\pi}{2} \right) \right] + 4 \left[\left(\frac{\sin \pi}{2} - \frac{\sin(-\pi)}{2} \right) \right]$$

$$\therefore A = 4 \left[\left(\frac{2\pi}{2} \right) \right] + 4 \left[\left(\frac{\sin \pi}{2} + \frac{\sin \pi}{2} \right) \right]$$

$$[\because \sin(-\theta) = -\sin \theta]$$

$$\therefore A = 4[\pi] + 4 \left[\left(\frac{0}{2} + \frac{0}{2} \right) \right]$$

$$[\because \sin \pi = 0]$$

$$\therefore A = 4[\pi] + 4.0$$

$$\therefore A = 4\pi \quad \text{Answer}$$

Justification

We know Area of a circle is πr^2

Here radius = $r = 2$

$$\therefore \text{Area of a circle is } \pi r^2 = \pi \times 2^2 = \pi \times 4 = 4\pi$$

 **Q # 96:** If $\vec{F} = 2y\hat{i} - z\hat{j} + x^2\hat{k}$ and S is the surface of the parabolic cylinder $y^2 = 8x$ in the first octant bounded by the planes $y = 4$ and $z = 6$. Evaluate $\iint_S \vec{F} \cdot \hat{\eta} \, ds$

Answer: Given, $\vec{F} = 2y\hat{i} - z\hat{j} + x^2\hat{k}$ and the scalar function $y^2 = 8x$ i.e.
 $y^2 - 8x = 0$ implies $8x - y^2 = 0$

Let the scalar function $\phi(x, y) = 8x - y^2$ of the given parabolic surface.

We have, $\vec{\nabla} \phi$ is normal (perpendicular) vector to the surface.

Given, $\phi(x, y) = 8x - y^2$

$$\begin{aligned} \therefore \vec{\nabla} \phi &= \left(\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \phi = \left(\frac{\delta \phi}{\delta x} \hat{i} + \frac{\delta \phi}{\delta y} \hat{j} + \frac{\delta \phi}{\delta z} \hat{k} \right) = \hat{i} \frac{\delta}{\delta x} \phi + \hat{j} \frac{\delta}{\delta y} \phi + \hat{k} \frac{\delta}{\delta z} \phi \\ &= \hat{i} \frac{\delta}{\delta x} (8x - y^2) + \hat{j} \frac{\delta}{\delta y} (8x - y^2) + \hat{k} \frac{\delta}{\delta z} (8x - y^2) \\ &= 8\hat{i} - 2y\hat{j} \end{aligned}$$

Let, $\hat{\eta}$ is the unit vector of $\vec{\nabla} \phi$

We can write,

$$\hat{\eta} = \frac{\vec{\nabla} \phi}{\left| \vec{\nabla} \phi \right|}$$

$$\hat{n} = \frac{8\hat{i} - 2y\hat{j}}{\sqrt{8^2 + (-2y)^2}} = \frac{8\hat{i} - 2y\hat{j}}{\sqrt{64 + 4y^2}} = \frac{2(4\hat{i} - y\hat{j})}{\sqrt{4(16 + y^2)}}$$

$$= \frac{2(4\hat{i} - y\hat{j})}{2\sqrt{(16 + y^2)}} = \frac{(4\hat{i} - y\hat{j})}{\sqrt{(16 + y^2)}}$$

Given, $\vec{F} = 2y\hat{i} - z\hat{j} + x^2\hat{k}$

$$\therefore \vec{F} \cdot \hat{n} = (2y\hat{i} - z\hat{j} + x^2\hat{k}) \cdot \frac{(4\hat{i} - y\hat{j})}{\sqrt{(16 + y^2)}}$$

$$\therefore \vec{F} \cdot \hat{n} = (2y\hat{i} - z\hat{j} + x^2\hat{k}) \cdot \frac{(4\hat{i} - y\hat{j} + 0\hat{k})}{\sqrt{(16 + y^2)}}$$

$$\therefore \vec{F} \cdot \hat{n} = \frac{(8y + yz)}{\sqrt{(16 + y^2)}}$$

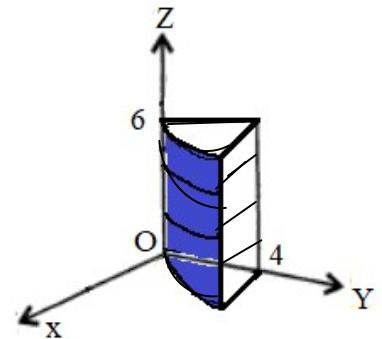
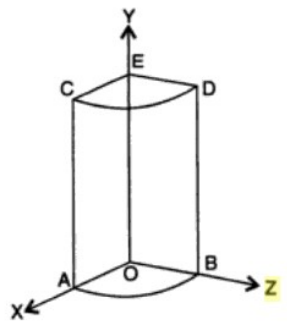
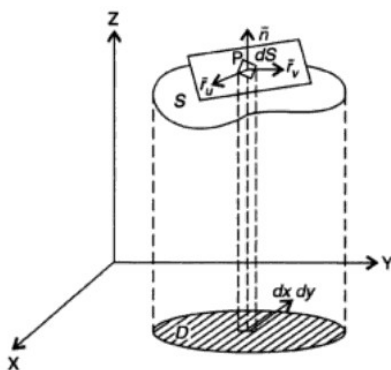
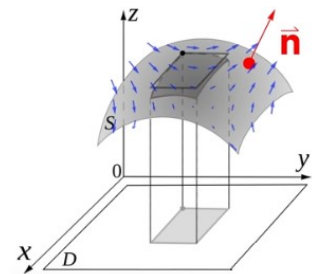
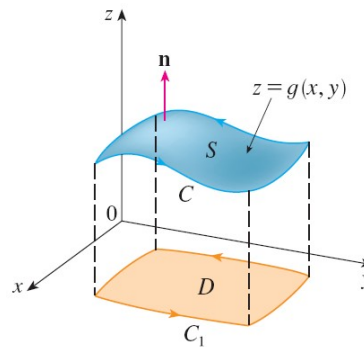
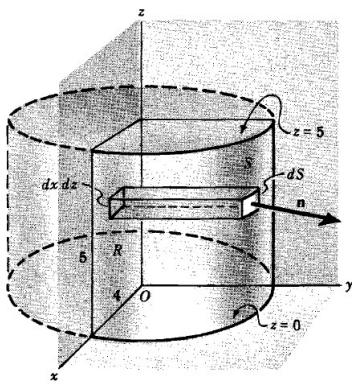


Figure # 128

A surface integral is evaluated by reducing it to a double integral by projecting the given surface S onto one of the coordinate planes. Let D be the projection of S onto the xy -

plane . Then $dS = \frac{dxdy}{\left| \hat{\eta} \cdot \hat{k} \right|}$; where $\hat{\eta}$ is the unit outward drawn normal to S. If D be the

projection of S onto the yz-plane . Then $dS = \frac{dydz}{\left| \hat{\eta} \cdot \hat{i} \right|}$;

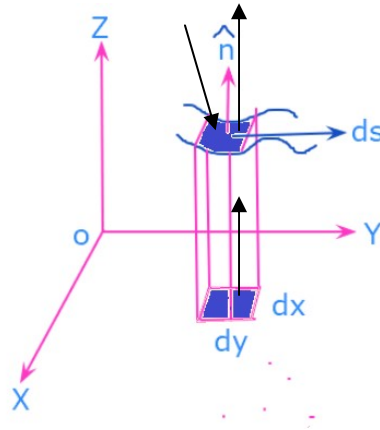


Figure # 129

$$\therefore \text{Projection of } \vec{A} \text{ on } \vec{B} \text{ is } = \frac{\vec{A} \cdot \vec{B}}{\left| \vec{B} \right|} \text{-----(i)}$$

Here, $\vec{A} = ds \hat{\eta}$; $\vec{B} = dxdy \hat{k}$; dxdy is the projection of ds

$$\text{From (i), } \frac{\vec{A} \cdot \vec{B}}{\left| \vec{B} \right|} = dxdy$$

$$\Rightarrow \frac{ds \hat{\eta} \cdot dxdy \hat{k}}{dxdy} = dxdy$$

$$\Rightarrow \frac{ds \left| \hat{\eta} \cdot \hat{k} \right|}{1} = dxdy$$

$$\Rightarrow ds = \frac{dxdy}{\left| \hat{\eta} \cdot \hat{k} \right|} \text{-----(ii)}$$

Similarly, If D be the projection of S onto the yz-plane . Then $dS = \frac{dydz}{\left| \hat{\eta} \cdot \hat{i} \right|}$;

We have, $\hat{\eta} = \frac{2(4\hat{i} - y\hat{j})}{2\sqrt{(16 + y^2)}} = \frac{(4\hat{i} - y\hat{j})}{\sqrt{(16 + y^2)}}$

$$\therefore \hat{\eta} \cdot \hat{i} = \frac{(4\hat{i} - y\hat{j})}{\sqrt{(16 + y^2)}} \cdot \hat{i} = \frac{4}{\sqrt{(16 + y^2)}}$$

$$\therefore \iint_s \vec{F} \cdot \hat{\eta} \, ds = \iint_s \vec{F} \cdot \hat{\eta} \frac{dydz}{\left| \hat{\eta} \cdot \hat{i} \right|}$$

$$\therefore \iint_s \vec{F} \cdot \hat{\eta} \, ds = \iint_s \frac{(8y + yz)}{\sqrt{(16 + y^2)}} \frac{dydz}{\left| \hat{\eta} \cdot \hat{i} \right|}$$

$$\therefore \iint_s \vec{F} \cdot \hat{\eta} \, ds = \iint_s \frac{(8y + yz)}{\sqrt{(16 + y^2)}} \frac{dydz}{\frac{4}{\sqrt{(16 + y^2)}}}$$

$$\therefore \iint_s \vec{F} \cdot \hat{\eta} \, ds = \iint_s \frac{(8y + yz)}{\sqrt{(16 + y^2)}} \frac{\sqrt{(16 + y^2)} dydz}{4}$$

$$\therefore \iint_s \vec{F} \cdot \hat{\eta} \, ds = \frac{1}{4} \iint_s (8y + yz) dydz$$

$$\therefore \iint_s \vec{F} \cdot \hat{\eta} \, ds = \frac{1}{4} \int_0^6 \int_0^4 (8y + yz) dydz$$

$$\therefore \iint_s \vec{F} \cdot \hat{\eta} \, ds = \frac{1}{4} \int_0^6 \int_0^4 (8 + z)y dydz = 132$$

Chapter 4: General Theorem

Divergence theorem (Gauss' theorem)

The relation between surface and volume integrals (Gauss divergence theorem)

For a closed surface S , enclosing a region V in a vector field $\vec{F}$.

$$\text{Then, } \int_V \text{div } \vec{F} dV = \int_S \vec{F} \cdot d\vec{S}$$

Q # 99: Verify the divergence theorem for the vector field $\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$ taken over the region bounded by the planes $z = 0, z = 2, x = 0, x = 1, y = 0, y = 3$

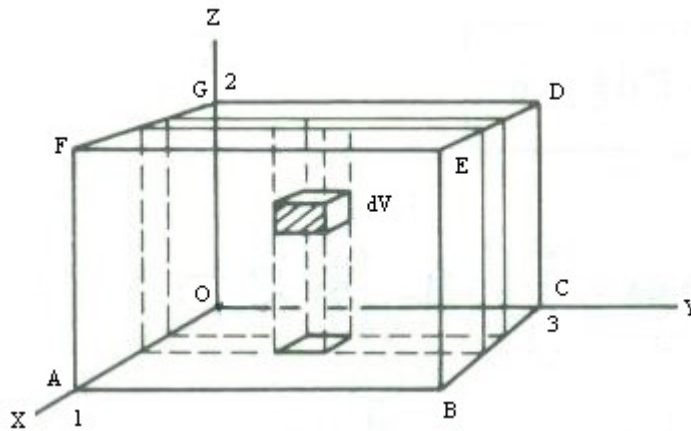


Figure # 132

Serial no	Surface	$\hat{\eta}=?$	ds	Plane
1	OABC	$\hat{\eta} = -\hat{k}$	$dx dy$	$z = 0$
2	DEFG	$\hat{\eta} = \hat{k}$	$dx dy$	$z = 2$
3	OAFG	$\hat{\eta} = -\hat{j}$	$dx dz$	$y = 0$
4	BCDE	$\hat{\eta} = \hat{j}$	$dx dz$	$y = 3$
5	OCDG	$\hat{\eta} = -\hat{i}$	$dy dz$	$x = 0$
6	ABEF	$\hat{\eta} = \hat{i}$	$dy dz$	$x = 1$

Answer:

$$dV = dx dy dz$$

$$\text{We have to show that } \int_V \text{div } \vec{F} dV = \int_S \vec{F} \cdot d\vec{S}$$

L.H.S.

Given, $\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$

$$\therefore \text{div } \vec{F} = \nabla \cdot \vec{F} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (x^2 \hat{i} + z \hat{j} + y \hat{k})$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \left(\frac{\partial}{\partial x} (x^2) + \frac{\partial}{\partial y} (z) + \frac{\partial}{\partial z} (y) \right)$$

$$= 2x + 0 + 0 = 2x$$

L.H.S.

$$\therefore \int_V \text{div } \vec{F} dV = \int_V 2x dV = \iiint_V 2x dz dy dx = \int_0^1 \int_0^3 \int_0^2 2x dz dy dx$$

$$= \int_0^1 \int_0^3 [2xz]_0^2 dy dx \quad \left[\int dz = z \right]$$

$$= \int_0^1 \int_0^3 [(2x \times 2) - (2x \times 0)] dy dx$$

$$= \int_0^1 \int_0^3 [4x] dy dx$$

$$= \int_0^1 [4xy]_0^3 dx \quad \left[\int dy = y \right]$$

$$= \int_0^1 [4x \times 3 - 4x \times 0] dx$$

$$= \int_0^1 [12x] dx$$

$$= \left[\frac{12x^2}{2} \right]_0^1 \quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} \right]$$

$$= [6x^2]_0^1 = [6 \times 1^2 - 6 \times 0^2] = 6$$

$$\therefore \int_V \text{div } \vec{F} dV = 6$$

R.H.S.

We have,

Any Vector = Length of this Vector $\times$ Unit Vector

$$\therefore d\vec{S} = \left| \frac{\vec{r}}{|\vec{r}|} \right| \hat{n}$$

or

$$\therefore d\vec{S} = dS \hat{n}$$

So, we can write

$$\int_S \vec{F} \cdot d\vec{S} = \int_S \vec{F} \cdot dS \hat{n}$$

The enclosing surface S consists of six separate plane faces denoted as

$S_1, S_2, S_3, S_4, S_5, S_6$ as shown. We consider each face in turn $\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$

(i) S_1 (OABC Base): $z = 0$; $\hat{n} = -\hat{k}$ (clockwise)

$$\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$$

$$\therefore \vec{F} = x^2 \hat{i} + 0 \hat{j} + y \hat{k} = x^2 \hat{i} + y \hat{k} \text{ and } dS_1 = dxdy \text{ [} dS = dxdy \text{]}$$

$$\int_{S_1} \vec{F} \cdot \hat{n} dS_1 = \iint_{S_1} (x^2 \hat{i} + y \hat{k}) \cdot (-\hat{k}) dy dx = \int_0^1 \int_0^3 (-y) dy dx$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \left[-\frac{y^2}{2} \right]_0^3 dx = \int_0^1 \left[-\frac{3^2}{2} + -\frac{0^2}{2} \right] dx = \int_0^1 \left[-\frac{9}{2} \right] dx = \left[-\frac{9}{2} x \right]_0^1 = -\frac{9}{2}(1-0) = -\frac{9}{2}$$

$$[\because \int dx = x]$$

ii) S_2 (DEFG Top): $z = 2$; $\hat{n} = \hat{k}$

$$\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$$

$$\therefore \vec{F} = x^2 \hat{i} + 2 \hat{j} + y \hat{k} \text{ and } dS_2 = dxdy$$

$$\int_{S_2} \vec{F} \cdot \hat{n} dS_2 = \iint_{S_2} (x^2 \hat{i} + 2 \hat{j} + y \hat{k}) \cdot (\hat{k}) dy dx = \int_0^1 \int_0^3 (y) dy dx$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \left[\frac{y^2}{2} \right]_0^3 dx = \int_0^1 \left[\frac{3^2}{2} - \frac{0^2}{2} \right] dx = \int_0^1 \left[\frac{9}{2} \right] dx = \left[\frac{9}{2} x \right]_0^1 = \frac{9}{2}(1-0) = \frac{9}{2} \quad [\because \int dx = x]$$

iii) S_3 (BCDE, Right-hand end): $y = 3$; $\hat{n} = \hat{j}$

$$\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$$

$$\therefore \vec{F} = x^2 \hat{i} + z \hat{j} + 3 \hat{k} \text{ and } dS_3 = dx dz$$

$$\int_{S_3} \vec{F} \cdot \hat{n} dS_3 = \iint_{S_3} (x^2 \hat{i} + z \hat{j} + 3 \hat{k}) \cdot (\hat{j}) dz dx = \int_0^1 \int_0^2 (z) dz dx$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \left[\frac{z^2}{2} \right]_0^2 dx = \int_0^1 \left[\frac{2^2}{2} - \frac{0^2}{2} \right] dx = \int_0^1 \left[\frac{4}{2} \right] dx = [2x]_0^1 = 2(1-0) = 2 \quad [\because \int dx = x]$$

$$\text{iv) } S_4 \text{ (OAFG, Left-hand end): } y = 0; \hat{n} = -\hat{j}$$

$$\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$$

$$\therefore \vec{F} = x^2 \hat{i} + z \hat{j} + 0 \hat{k} \text{ and } dS_4 = dx dz$$

$$\int_{S_4} \vec{F} \cdot \hat{n} dS_4 = \iint_{S_4} (x^2 \hat{i} + z \hat{j} + 0 \hat{k}) \cdot (-\hat{j}) dz dx = \int_0^1 \int_0^2 (-z) dz dx$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^1 \left[-\frac{z^2}{2} \right]_0^2 dx = \int_0^1 \left[-\frac{2^2}{2} + -\frac{0^2}{2} \right] dx = \int_0^1 \left[-\frac{4}{2} \right] dx = [-2x]_0^1 = -2(1-0) = -2 \quad [\because \int dx = x]$$

$$\text{v) } S_5 \text{ (ABEF, Front): } x = 1; \hat{n} = \hat{i}$$

$$\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$$

$$\therefore \vec{F} = 1^2 \hat{i} + z \hat{j} + y \hat{k} \text{ and } dS_5 = dy dz$$

$$\int_{S_5} \vec{F} \cdot \hat{n} dS_5 = \iint_{S_5} (1^2 \hat{i} + z \hat{j} + y \hat{k}) \cdot (\hat{i}) dy dz = \int_0^2 \int_0^3 dy dz$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

$$= \int_0^2 [y]_0^3 dz = \int_0^2 [3-0] dz = \int_0^2 [3] dz = [3z]_0^2 = 3(2-0) = 6 \quad [\because \int dy = y]$$

$$\text{vi) } S_6 \text{ (OCDG, Back): } x = 0; \hat{n} = -\hat{i}$$

$$\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$$

$$\therefore \vec{F} = 0 \hat{i} + z \hat{j} + y \hat{k} \text{ and } dS_6 = dy dz$$

$$\int_{S_6} \vec{F} \cdot \hat{n} dS_6 = \iint_{S_6} (0 \hat{i} + z \hat{j} + y \hat{k}) \cdot (-\hat{i}) dy dz = 0$$

For the whole surface S we therefore have

$$\int_s \vec{F} \cdot d\vec{S} = -\frac{9}{2} + \frac{9}{2} + 2 - 2 + 6 + 0 = 6$$

Q # 100: Home task:

Verify the Divergence theorem for $\vec{F} = (2xy + z)\hat{i} + y^2\hat{j} - (x + 3y)\hat{k}$ taken over the region bounded by the planes, $2x + 2y + z = 6, x = 0, y = 0, z = 0$

Hints;

When, $x = 0, y = 0$ then $2x + 2y + z = 6$

$$2.0 + 2.0 + z = 6$$

$$z = 6$$

When, $y = 0, z = 0$ then $2x + 2y + z = 6$

$$2x + 2.0 + 0 = 6$$

$$2x = 6$$

$$x = 3$$

When, $x = 0, z = 0$ then $2x + 2y + z = 6$

$$2.0 + 2y + 0 = 6$$

$$2y = 6$$

$$y = 3$$

So, the above problem can be written as

Verify the Divergence theorem for $\vec{F} = (2xy + z)\hat{i} + y^2\hat{j} - (x + 3y)\hat{k}$ taken over the region bounded by the planes, $z = 0, z = 6, x = 0, x = 3, y = 0, y = 3$

Q # 101: Evaluate $I = \oint_C (3x^2y^2dx + 2x^3ydy)$ between $O(0,0)$ and $A(2,4)$

a) along c_1 i.e. $y = x^2$

b) along c_2 i.e. $y = 2x$

c) along c_3 i.e. $x = 0$ from $(0,0)$ to $(0,4)$ and $y = 4$ from $(0,4)$ to $(2,4)$

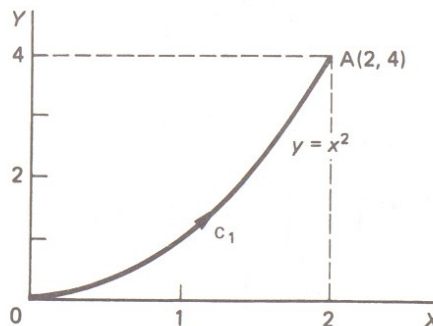


Figure # 133

$$I = \oint_C (3x^2y^2dx + 2x^3ydy)$$

The path c_1 is $y = x^2$

We have, Green's theorem

$$\therefore \oint_C (Pdx + Qdy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

$$\oint_C (Pdx + Qdy) = - \iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) dx dy$$

Now,

$$I = - \iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) dx dy$$

$$I = - \iint_R (1 - 3) dx dy$$

$$I = - \iint_R -2 dx dy$$

$$I = 2 \iint_R dx dy = 2A \quad \text{-----(i)}$$

$$\left[\iint_R dx dy = A \right]$$

Now, The area of the triangle: $A = \frac{1}{2} \times \text{base} \times \text{height}$

$$A = \frac{1}{2} \times OA \times AB \quad \text{[Figure 122]}$$

$$A = \frac{1}{2} \times 1 \times 2 = 1$$

From (i),

$$I = 2 \times A = 2 \times 1 = 2 \quad [A = 1]$$

L.H.S = R.H.S (Proved)

Q# 104: Evaluate the line Integral $I = \oint_C \{xydx + (2x - y)dy\}$ round the region bounded by

the curves $y = x^2$ and $x = y^2$ by the use of Green's theorem.

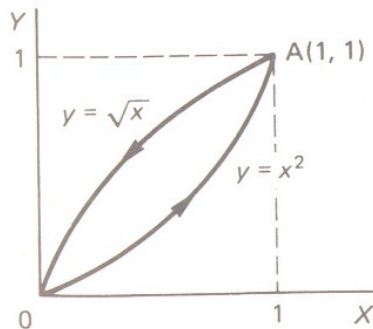


Figure # 141

Answer: Points of intersection are $O(0,0)$ and $A(1,1)$. P and Q are known, so there is no difficulty.

Now, By Green's theorem

$$I = \oint_C \{xydx + (2x - y)dy\}$$

Here, $P = xy$

$$\therefore \frac{\partial P}{\partial y} = x$$

and $Q = 2x - y$

$$\therefore \frac{\partial Q}{\partial x} = 2$$

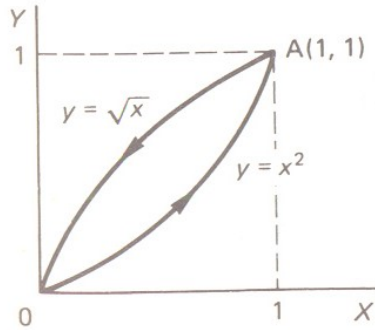


Figure # 142

$$\text{We have, } \iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) dx dy = - \oint_C (Pdx + Qdy)$$

$$\therefore - \iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) dx dy = \oint_C (Pdx + Qdy)$$

$$\therefore \oint_C (Pdx + Qdy) = - \iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) dx dy$$

$$\text{Now, } I = - \iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) dx dy$$

$$I = - \iint_R (x - 2) dx dy$$

$$I = - \int_0^1 \int_{y=x^2}^{y=\sqrt{x}} (x - 2) dy dx \quad [\text{Here upper limit of } y \text{ is } y = \sqrt{x} \text{ and the lower limit is } y = x^2]$$

$$I = - \int_0^1 (x - 2) [y]_{x^2}^{\sqrt{x}} dx$$

$$\therefore I = - \int_0^1 (x - 2) [\sqrt{x} - x^2] dx$$

$$\therefore I = - \int_0^1 (x^{3/2} - x^3 - 2x^{1/2} + 2x^2) dx$$

$$\therefore I = - \left[\frac{2}{5} x^{5/2} - \frac{1}{4} x^4 - \frac{4}{3} x^{3/2} + \frac{2}{3} x^3 \right]_0^1 = \frac{31}{60}$$

Now, L.H.S. $\oint Pdx + Qdy$

$$= \oint xydx + (2x - y)dy$$

$$I_1 = \int_0^1 x \cdot x^2 dx + (2x - x^2) 2x dx$$

$$= \int_0^1 x^3 dx + 4x^2 - 2x^3 dx$$

$$= \int_0^1 (-x^3 dx + 4x^2) dx$$

$$= \left[\frac{-x^4}{4} + 4 \frac{x^3}{3} \right]_0^1$$

$$= \left[\frac{-1}{4} + \frac{4}{3} \right] = \frac{13}{12}$$

and $\oint Pdx + Qdy$

$$= \oint xydx + (2x - y)dy$$

$$I_2 = \int_1^0 xydx + (2x - y)dy$$

$$I_2 = \int_1^0 x \cdot \sqrt{x} dx + (2x - \sqrt{x}) \frac{1}{2} x^{-1/2} dx$$

$$I_2 = \int_1^0 (x^{3/2} + x^{1/2} - \frac{1}{2}) dx$$

$$I_2 = \left[\frac{x^{5/2}}{\frac{5}{2}} + \frac{x^{3/2}}{\frac{3}{2}} - \frac{1}{2} x \right]_1^0$$

$$I_2 = \left[0 - \frac{2}{5} - \frac{2}{3} + \frac{1}{2} \right] = -\frac{17}{30}$$

$$\text{Finally } I = I_1 + I_2 = \frac{13}{12} - \frac{17}{30} = \frac{31}{60}$$

L.H.S = R.H.S (Proved)

$$\begin{aligned}
&= \int_{-1}^1 \left[\frac{1^2}{2} - \frac{(-1)^2}{2} \right] dy \\
&= \int_{-1}^1 \left[\frac{1^2}{2} - \frac{1^2}{2} \right] dy \\
&= 0 \text{ Answer}
\end{aligned}$$

Q# 112: Home Task:

Verify Green's theorem in the plane for $\int_C (2xy - x^2)dx + (x + y^2)dy$ where the close curve of the region bounded by $y = x, y^2 = x$

